DUALITY IN DECONFOUNDING WITH INSTRUMENTS AND SURROGATE CONTROLS

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Abstract: Instrumental variables and surrogate controls address unmeasured confounding through different sources of variation. We develop a unified framework for linear structural models that characterizes their identifying information and the gains from combining them. The framework uses two projected moment restrictions: one based on the component of the regressors explained by instruments, and the other on the component orthogonal to surrogate controls. We characterize the statistical targets, identification conditions, and asymptotic distributions of the corresponding estimators. Combining the two moment blocks through generalized method of moments yields complementary identification, allowing joint identification even when neither block alone identifies the full coefficient vector. We characterize when projection preserves identification and efficiency, and establish the efficiency gains from optimal combination under correct specification. Under local misspecification, we derive the asymptotic bias and variance and develop feasible weights that minimize worst-case asymptotic squared-error risk over a prespecified ellipsoidal set of moment violations. Simu-

lations and an application to single-cell CRISPR perturbation data illustrate the efficiency gains and sensitivity to moment misspecification.

Key words and phrases: Generalized method of moments, instrumental variables, local misspecification, surrogate controls.

1. Introduction

Designed perturbations and auxiliary measurements provide opportunities to address unmeasured confounding. However, randomization of a perturbation does not ensure exogeneity of the resulting exposure. We study how instruments and surrogate controls can be used separately and jointly to estimate linear structural effects.

Single-cell Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) perturbation studies provide a motivating example. For a fixed target gene g, let z_i denote the assigned single-guide RNA (sgRNA) or guide-pair indicator, x_i the measured expression of the target gene, and y_i the expression of a candidate downstream gene. Latent cellular and technical factors may confound the relationship between x_i and y_i, despite experimental assignment of z_i. The assignment z_i provides a candidate instrument, while surrogate controls w_i, constructed for example from negative-control or housekeeping genes, may provide information about la-

tent confounding. These sources are complementary: instruments may be low-dimensional or weak, whereas surrogate controls may provide richer confounding information.

We study a linear framework in which instrumental-variable (IV) and surrogate-control (SV) methods generate two projected moment restrictions for the same structural parameter. In the scalar-outcome model $y_i = x_i^\top \beta + \varsigma_i$, the IV channel uses $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$, implied by $\mathbb{E}(z_i\varsigma_i) = 0_q$, whereas the SV channel uses $\mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = 0_p$. Geometrically, IV retains the component of x_i explained by z_i, whereas SV retains the component orthogonal to the span of w_i. Both yield p-dimensional projected moment blocks, providing a common framework for identification, first-order behavior, joint weighting, and sensitivity to misspecification. The construction extends columnwise to multivariate outcomes $Y = XB + \Xi$.

Our framework connects several strands of the literature. Classical instrumental-variable methods address endogeneity through relevance and orthogonality restrictions, with 2SLS and GMM providing standard projection- and moment-based approaches (Anderson and Rubin, 1949; Theil, 1953; Sargan, 1958; Hansen, 1982). Our projected IV formulation uses the p-dimensional component of the classical q-dimensional IV moment relevant for projecting x_i. Related proxy-variable, negative-control, proximal, and

latent-factor methods use auxiliary measurements to address latent confounding (Miao et al., 2018; Shi et al., 2020; Miao et al., 2024; Tchetgen Tchetgen et al., 2024; Wang and Blei, 2019; Adams and Hansen, 2025). Particularly close is Miao et al. (2023), which unifies IV-like and proxy-like information in a nonparametric identification framework; we instead focus on a linear structural coefficient and the projection, rank, first-order, and weighting structure of the resulting moments. Our framework is also related to recent work combining multiple deconfounding signals (Dukes et al., 2025; Imbens et al., 2025; Kallus and Mao, 2025). Rather than recovering the latent confounder, we treat z_i and w_i as observed ancillary signals and study their projected moments under exact and locally misspecified restrictions.

Our contributions are threefold. First, we place IV and SV adjustment in a common p-dimensional projected-moment representation based on $\Pi_z x_i$ and $\Pi_w^{\perp} x_i$, respectively, making explicit their shared structure and distinct projection geometries. Second, for each single-source channel, we characterize the statistical target under projected-moment violations and establish identification, consistency, and asymptotic normality of the corresponding feasible estimator, while separating statistical identification from the additional assumptions required for causal interpretation (Theorems 1 and 2).

Third, we combine the two projected blocks in a $2p$-dimensional GMM system and characterize complementary identification through the stacked Jacobian, which can identify the coefficient even when an individual block is rank deficient. We establish the corresponding first-order theory under both exact and $n^{-1/2}$-local misspecification (Proposition 3 and theorems 4 and 5). Building on this joint system, we develop risk-optimal weighting that trades sensitivity to local moment violations against variance, establish a feasible implementation, and compare its risk with the single-source procedures under exact and locally misspecified regimes (Lemma 6 and propositions 7–9). We also characterize the identifying information retained by IV projection and its efficiency relative to the raw IV moments (Propositions 11 and 12).

2. Preliminaries

2.1 Auxiliary variables for deconfounding

Consider the confounded linear regression model

$$y = x^\top \beta + \varsigma, \tag{2.1}$$

where $y \in \mathbb{R}$, $x \in \mathbb{R}^p$, $\beta \in \mathbb{R}^p$, and $\varsigma \in \mathbb{R}$. We assume $\mathbb{E}(\varsigma) = 0$, $\mathrm{Var}(\varsigma) = \sigma^2 < \infty$, and $\Sigma_x := \mathbb{E}(xx^\top) \in \mathbb{S}^p_{++}$. When $\mathbb{E}(x\varsigma) = 0$, ordinary least squares consistently estimates β. Our interest is instead in settings

where $\mathbb{E}(x\varsigma) \neq 0$, for example because the disturbance contains a latent confounding component $u \in \mathbb{R}^r$ associated with both the regressors and the outcome; see Figure 1.

For n independent and identically distributed (i.i.d.) observations, write the sample model as

$$Y = X\beta + \varsigma, \tag{2.2}$$

where $Y = (y_1, \ldots, y_n)^\top \in \mathbb{R}^n$, $X = (x_1, \ldots, x_n)^\top \in \mathbb{R}^{n \times p}$, and $\varsigma = (\varsigma_1, \ldots, \varsigma_n)^\top \in \mathbb{R}^n$.

To address latent confounding, we use observed auxiliary variables that provide information beyond x_i. The IV and SV channels exploit, respectively, instrument-induced and residualized variation in x_i. Their distinct projection geometries are illustrated in Figure 1 and developed below.

2.2 Instrumental variable models

Suppose, in addition, that an instrument $z_i \in \mathbb{R}^q$ is observed for each i. Under the centered formulation, let $\Sigma_z := \mathbb{E}(z_i z_i^\top)$, $\Sigma_{xz} := \mathbb{E}(x_i z_i^\top)$, and $\Sigma_{zx} := \Sigma_{xz}^\top$. When $\Sigma_z \in \mathbb{S}_{++}^q$, define the population projection $\Pi_z x_i := \Sigma_{xz}\Sigma_z^{-1} z_i$.

Define the projected-moment bias $\delta_{\text{IV}} := \mathbb{E}\{(\Pi_z x_i)\varsigma_i\}$ and relevance matrix $Q_{\text{IV}} := \mathbb{E}\{(\Pi_z x_i)x_i^\top\} = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$. Whenever $Q_{\text{IV}} \in \mathbb{S}_{++}^p$, the corre-

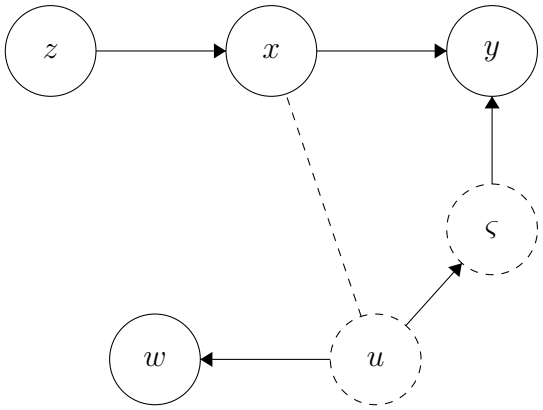


Figure 1: Illustrative causal diagram for instrumental variables and surrogate controls. The diagram imposes stronger causal restrictions than the projected moment conditions used in the analysis. Solid and dashed circles denote observed and unobserved variables, respectively.

sponding statistical target is $\beta_{\text{IV}} := \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}}$. Thus, the raw IV restriction $\mathbb{E}(z_i\varsigma_i) = 0_q$ implies $\delta_{\text{IV}} = 0_p$ and $\beta_{\text{IV}} = \beta$. A nonzero δ_{IV} represents misspecification of the projected IV moment.

At the sample level, let $Z = (z_1, \ldots, z_n)^\top \in \mathbb{R}^{n\times q}$ and $P_Z := Z(Z^\top Z)^{-1}Z^\top$. The first stage projects the regressors onto the instrument span, giving the fitted design $\widehat{X} := P_Z X$. Whenever $Z^\top Z$ and $X^\top P_Z X$ are nonsingular, the second-stage estimator is

$$\widehat{\beta}_{\text{IV}} := \operatorname*{argmin}_{\theta\in\mathbb{R}^p} \|Y - \widehat{X}\theta\|_2^2 = (X^\top P_Z X)^{-1}X^\top P_Z Y. \tag{2.3}$$

Equivalently, the sample rank conditions are $\operatorname{rank}(Z) = q$ and $\operatorname{rank}(Z^\top X) =$

p, which necessarily require $p \leq q \leq n$. Throughout the fixed-dimensional asymptotic analysis, p and q are treated as fixed as $n \to \infty$.

For the asymptotic distribution, define $e_{\mathrm{IV},i} := y_i - x_i^\top \beta_{\mathrm{IV}} - z_i^\top \Sigma_z^{-1} \mathbb{E}\{z_i(y_i - x_i^\top \beta_{\mathrm{IV}})\}$. Then $\mathbb{E}(z_i e_{\mathrm{IV},i}) = 0_q$. The first-order score is $\psi_{\mathrm{IV},i} := (\Pi_z x_i) e_{\mathrm{IV},i} + x_i z_i^\top \Sigma_z^{-1} \mathbb{E}\{z_i(y_i - x_i^\top \beta_{\mathrm{IV}})\}$, with covariance $\Sigma_{\mathrm{IV}} := \mathrm{Var}(\psi_{\mathrm{IV},i})$.

Assumption 1 (Projected instrumental variables conditions)**.** Suppose the following conditions hold.

(i) *Projected relevance.* Suppose $\Sigma_z \in \mathbb{S}_{++}^q$ and $Q_{\mathrm{IV}} \in \mathbb{S}_{++}^p$.

(ii) *Second moments and nondegeneracy.* Suppose $\mathbb{E}[\|x_i z_i^\top\|_{\mathrm{op}}^2] < \infty$, $\mathbb{E}\|z_i e_{\mathrm{IV},i}\|_2^2 < \infty$, and $\Sigma_{\mathrm{IV}} \in \mathbb{S}_{++}^p$.

Theorem 1 (Identification and asymptotic properties of projected 2SLS)**.** Under Assumption 1(i), the following statements hold. (i) *Identification.* The projected population moment equation $\mathbb{E}[(\Pi_z x_i)(y_i - x_i^\top \theta)] = 0$ has the unique solution $\theta = \beta_{\mathrm{IV}}$. (ii) *Consistency.* As $n \to \infty$, $\widehat{\beta}_{\mathrm{IV}} \xrightarrow{\mathrm{p}} \beta_{\mathrm{IV}}$. (iii) *Asymptotic normality.* If, in addition, Assumption 1(ii) holds, then $\sqrt{n}(\widehat{\beta}_{\mathrm{IV}} - \beta_{\mathrm{IV}}) \xrightarrow{\mathrm{d}} \mathcal{N}(0, V_{\mathrm{IV}})$, where $V_{\mathrm{IV}} := Q_{\mathrm{IV}}^{-1} \Sigma_{\mathrm{IV}} Q_{\mathrm{IV}}^{-1}$.

Moreover, $\mathrm{rank}(Q_{\mathrm{IV}}) = \mathrm{rank}(\Sigma_{zx}) \leq \min(p, q)$, so IV-only identification requires $\mathrm{rank}(\Sigma_{zx}) = p$ and hence $q \geq p$, although the stacked IV–SV system may remain identifying when an individual block is rank deficient.

When $q > p$, projection may discard overidentifying information; see Appendix A.1 for further discussion.

2.3 Surrogate controls and partialling out

Suppose, in addition, that surrogate controls $w_i \in \mathbb{R}^m$ are observed. Under the centered formulation, let $\Sigma_w := \mathbb{E}(w_i w_i^\top)$ and $\Sigma_{xw} := \mathbb{E}(x_i w_i^\top)$. When $\Sigma_w \in \mathbb{S}^m_{++}$, define $\Pi_w x_i := \Sigma_{xw}\Sigma_w^{-1} w_i$ and $\Pi_w^\perp x_i := x_i - \Pi_w x_i$. Unlike IV, SV retains the component of x_i orthogonal to the surrogate-control span. Exact validity requires $\mathbb{E}[(\Pi_w^\perp x_i)\varsigma_i] = 0_p$, without requiring $\mathbb{E}(x_i\varsigma_i) = 0_p$ or exogeneity of w_i.

Define the residualized-moment bias $\delta_{\mathrm{sv}} := \mathbb{E}[(\Pi_w^\perp x_i)\varsigma_i]$ and relevance matrix $Q_{\mathrm{sv}} := \mathbb{E}[(\Pi_w^\perp x_i)x_i^\top]$. Whenever $Q_{\mathrm{sv}} \in \mathbb{S}^p_{++}$, the corresponding statistical target is $\beta_{\mathrm{sv}} := \beta + Q_{\mathrm{sv}}^{-1}\delta_{\mathrm{sv}}$. Hence, exact validity gives $\beta_{\mathrm{sv}} = \beta$, while $\delta_{\mathrm{sv}} \neq 0_p$ represents residual confounding after surrogate adjustment.

At the sample level, let $W = (w_1, \ldots, w_n)^\top \in \mathbb{R}^{n\times m}$, $P_W := W(W^\top W)^{-1}W^\top$, and $P_W^\perp := I_n - P_W$. The first step partials out the surrogate controls from both the regressors and the outcome, giving $\widetilde{X} := P_W^\perp X$ and $\widetilde{Y} := P_W^\perp Y$. When $W^\top W$ and $X^\top P_W^\perp X$ are nonsingular, the second-step estimator is

$$\widehat{\beta}_{\mathrm{sv}} := \operatorname*{argmin}_{\theta\in\mathbb{R}^p} \|\widetilde{Y} - \widetilde{X}\theta\|_2^2 = (X^\top P_W^\perp X)^{-1} X^\top P_W^\perp Y. \tag{2.4}$$

Equivalently, the sample rank conditions are $\operatorname{rank}(W) = m$ and $\operatorname{rank}(P_W^\perp X) =$

p, requiring $m + p \leq n$; throughout, m and p are fixed as $n \to \infty$.

For the asymptotic distribution, define $e_{\mathrm{SV},i} := y_i - x_i^\top \beta_{\mathrm{SV}} - w_i^\top \Sigma_w^{-1} \mathbb{E}\{w_i(y_i - x_i^\top \beta_{\mathrm{SV}})\}$. Then $\mathbb{E}(w_i e_{\mathrm{SV},i}) = 0_m$. The first-order score is $\psi_{\mathrm{SV},i} := (\Pi_w^\perp x_i) e_{\mathrm{SV},i}$, with covariance $\Sigma_{\mathrm{SV}} := \mathrm{Var}(\psi_{\mathrm{SV},i})$.

Assumption 2 (Partialling-out conditions with observed controls)**.** Suppose the following conditions hold.

(i) *Residualized relevance.* Suppose $\Sigma_w \in \mathbb{S}_{++}^m$ and $Q_{\mathrm{SV}} \in \mathbb{S}_{++}^p$.

(ii) *Second moments and nondegeneracy.* Suppose $\mathbb{E}\|x_i \varsigma_i\|_2^2 < \infty$, $\mathbb{E}\|x_i x_i^\top\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|w_i e_{\mathrm{SV},i}\|_2^2 < \infty$, and $\Sigma_{\mathrm{SV}} \in \mathbb{S}_{++}^p$.

Theorem 2 (Identification and asymptotic properties of partialling out)**.** Under Assumption 2(i), the following statements hold. (i) *Identification.* The residualized population moment equation $\mathbb{E}[(\Pi_w^\perp x_i)(y_i - x_i^\top \theta)] = 0$ has the unique solution $\theta = \beta_{\mathrm{SV}}$. (ii) *Consistency.* As $n \to \infty$, $\widehat{\beta}_{\mathrm{SV}} \xrightarrow{\mathrm{p}} \beta_{\mathrm{SV}}$. (iii) *Asymptotic normality.* If, in addition, Assumption 2(ii) holds, then $\sqrt{n}(\widehat{\beta}_{\mathrm{SV}} - \beta_{\mathrm{SV}}) \xrightarrow{\mathrm{d}} \mathcal{N}(0, V_{\mathrm{SV}})$, where $V_{\mathrm{SV}} := Q_{\mathrm{SV}}^{-1} \Sigma_{\mathrm{SV}} Q_{\mathrm{SV}}^{-1}$.

Under exact surrogate-control validity, $\delta_{\mathrm{SV}} = 0_p$, we have $\beta_{\mathrm{SV}} = \beta$ and hence $\widehat{\beta}_{\mathrm{SV}} \xrightarrow{\mathrm{p}} \beta$. Identification requires sufficiently rich variation in x_i to remain after partialling out the surrogate controls; see Appendix A.2 for further discussion of the residualization geometry.

2.4 Causal interpretation

Theorems 1 and 2 identify the statistical targets $\beta_{\text{IV}} = \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}}$ and $\beta_{\text{SV}} = \beta + Q_{\text{SV}}^{-1}\delta_{\text{SV}}$. A causal interpretation as the structural coefficient β therefore requires the corresponding projected bias to vanish. For IV, consistency, exclusion, and exogeneity, together with the homogeneous linear response $y_i(x) = \alpha_i + \beta^\top x$, imply $\mathbb{E}(z_i\varsigma_i) = 0_q$, hence $\delta_{\text{IV}} = 0_p$ and $\beta_{\text{IV}} = \beta$ (Wooldridge, 2010); under heterogeneous effects, the linear IV coefficient need not represent a common causal effect and may instead have a local interpretation under additional conditions (Imbens and Angrist, 1994). For SV, causal validity requires the surrogate controls to capture the confounding-relevant component of x_i, so that $\mathbb{E}[(\Pi_w^\perp x_i)\varsigma_i] = 0_p$, implying $\beta_{\text{SV}} = \beta$. Thus, IV uses instrument-selected variation whereas SV uses residualized variation after surrogate adjustment; when the corresponding causal assumptions fail, the estimators remain well defined but generally target β_{IV} and β_{SV} rather than β.

3. Combining instruments and surrogate controls

The IV and SV approaches yield two projected moment restrictions for the same structural coefficient β. We combine these restrictions in a projected GMM system, characterize complementary identification, and derive the es-

timator's first-order behavior under exact and $n^{-1/2}$-local misspecification.

3.1 Projected GMM formulation and complementary identification

Consider the structural model (2.2). Under exact projected validity, the two moment restrictions are

$$\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p, \qquad \mathbb{E}\{(\Pi_w^{\perp} x_i)\varsigma_i\} = 0_p.$$

Define the stacked projected regressors and their cross moment with x_i by

$$h_i := \begin{pmatrix} \Pi_z x_i \\ \Pi_w^{\perp} x_i \end{pmatrix} \in \mathbb{R}^{2p}, \qquad J := \mathbb{E}(h_i x_i^{\top}) = \begin{pmatrix} Q_{\text{IV}} \\ Q_{\text{SV}} \end{pmatrix}.$$

The population moment has Jacobian $-J$ with respect to the coefficient vector. The joint restriction is

$$\mathbb{E}\{h_i(y_i - x_i^{\top}\beta)\} = 0_{2p}. \tag{3.1}$$

Proposition 3 (Complementary identification)**.** Suppose that (3.1) holds. Then β is uniquely identified by the stacked projected system if and only if

$$\operatorname{rank}(J) = p \quad \Longleftrightarrow \quad \ker(Q_{\text{IV}}) \cap \ker(Q_{\text{SV}}) = \{0_p\}. \tag{3.2}$$

Neither block needs to identify the full coefficient separately, provided that their unidentified directions have trivial intersection.

Indeed, equality of the population moments at θ and β is equivalent to $J(\theta - \beta) = 0_{2p}$. Hence a coefficient direction is unidentified by the joint system only if it is invisible to both projected moment blocks. The condition $\ker(Q_{\text{IV}}) \cap \ker(Q_{\text{SV}}) = \{0_p\}$ therefore formalizes complementarity: a direction not identified by one source may still be identified by the other. Joint identification can consequently hold even when both Q_{IV} and Q_{SV} are individually rank deficient, provided that their rank deficiencies occur along different directions.

This distinction separates complementary identification from the more familiar role of additional moments in improving precision. When both blocks are individually full rank, stacking primarily provides additional information for estimation and weighting. When one or both blocks are rank deficient, however, stacking can enlarge the identified subspace itself. Thus, the two ancillary sources need not be redundant: their value may arise from identifying different components of the same structural coefficient.

For the IV block, $\text{rank}(Q_{\text{IV}}) = \text{rank}(\Sigma_{zx}) \leq q$, so $q < p$ precludes IV-only identification but does not preclude identification by the stacked system. In this case, directions not recovered from the instrument-induced component of x_i may still be identified through variation remaining after projection on the surrogate controls. Conversely, directions weakly repre-

sented in the SV residualized variation may be supplied by the IV block. The identifying content of the SV block is determined by $\operatorname{rank}(Q_{\mathrm{SV}})$, without an analogous requirement $m \geq p$.

Define the sample moment $\widehat{g}(\theta) := n^{-1}\{[X^\top P_Z(Y - X\theta)]^\top, [X^\top P_W^\perp(Y - X\theta)]^\top\}^\top$ and its negative Jacobian $\widehat{J} := n^{-1}\{(X^\top P_Z X)^\top, (X^\top P_W^\perp X)^\top\}^\top$. For a fixed symmetric weight $M \succeq 0$, define

$$\widehat{\beta}(M) := \operatorname*{argmin}_{\theta \in \mathbb{R}^p} \widehat{g}(\theta)^\top M \widehat{g}(\theta). \tag{3.3}$$

Whenever $\widehat{J}^\top M \widehat{J}$ is nonsingular, the minimizer is unique and satisfies

$$\widehat{\beta}(M) = (\widehat{J}^\top M \widehat{J})^{-1} \widehat{J}^\top M \frac{1}{n} \begin{pmatrix} X^\top P_Z Y \\ X^\top P_W^\perp Y \end{pmatrix}. \tag{3.4}$$

For $M \in \mathbb{S}_{++}^{2p}$, population uniqueness is equivalent to $\operatorname{rank}(J) = p$.

If the relevant sample block is nonsingular, the semidefinite weights $\operatorname{diag}(I_p, 0_{p\times p})$ and $\operatorname{diag}(0_{p\times p}, I_p)$ recover $\widehat{\beta}_{\mathrm{IV}}$ and $\widehat{\beta}_{\mathrm{SV}}$, respectively. Section 4 also characterizes positive-definite weights inducing their population sensitivities. When $q > p$, projection may discard overidentifying information from the raw IV moment; Appendix D provides the corresponding efficiency comparison.

3.2 Asymptotic theory under exact specification and local misspecification

Because the sample moments use estimated projection operators, their first-order terms generally differ from $h_i\varsigma_i$. Define the terms at the structural coefficient β by

$$\psi_{\mathrm{IV},i} = (\Pi_z x_i)\varsigma_i + (x_i - \Pi_z x_i) z_i^\top \Sigma_z^{-1} \mathbb{E}(z_i\varsigma_i),$$
$$\psi_{\mathrm{SV},i} = (\Pi_w^\perp x_i)\{\varsigma_i - w_i^\top \Sigma_w^{-1} \mathbb{E}(w_i\varsigma_i)\}. \tag{3.5}$$

When an individual projected system is identified and valid, its statistical target equals β, and the corresponding term agrees with the single-source score defined earlier. These definitions remain applicable when an individual block is rank deficient.

Under the raw IV restriction $\mathbb{E}(z_i\varsigma_i) = 0_q$, the IV projection correction vanishes. The SV term involves the disturbance residualized on w_i, reflecting estimation of the surrogate projection. Write $\psi_i := (\psi_{\mathrm{IV},i}^\top, \psi_{\mathrm{SV},i}^\top)^\top$, $\Sigma := \mathrm{Var}(\psi_i)$, $K_M := (J^\top M J)^{-1} J^\top M$, and $V(M) := K_M \Sigma K_M^\top$.

Assumption 3 (Combined GMM conditions)**.** Suppose that the observations are i.i.d., the dimensions p, q, m are fixed, and the maintained second-moment conditions of Section 2 hold.

(i) **Projected validity and identification.** The restrictions in (3.1)

hold, $\Sigma_z \in \mathbb{S}^q_{++}$, and $\Sigma_w \in \mathbb{S}^m_{++}$. The fixed symmetric weight $M \succeq 0$ satisfies $J^\top M J \in \mathbb{S}^p_{++}$.

(ii) **Score moments and nondegeneracy.** Suppose $\mathbb{E}\|x_i z_i^\top\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|z_i\{\varsigma_i - z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)\}\|_2^2 < \infty$, $\mathbb{E}\|x_i\varsigma_i\|_2^2 < \infty$, $\mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty$, and $\mathbb{E}\|w_i\{\varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i)\}\|_2^2 < \infty$. Further, $V(M) \in \mathbb{S}^p_{++}$.

The covariance Σ may be singular because the system can contain redundant score directions. Positive definiteness of $V(M)$ ensures nondegenerate coefficient inference. Positive definiteness of Σ is required when its inverse is used for efficient weighting.

Theorem 4 (Identification, consistency, and asymptotic normality under exact specification)**.** Under Assumption 3 (i), the following statements hold. (i) β is the unique solution to the population stacked moment restriction. (ii) $\widehat{\beta}(M) \xrightarrow{\mathrm{p}} \beta$. (iii) If Assumption 3 (ii) also holds, then $\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\mathrm{d}} \mathcal{N}\{0, V(M)\}$.

We next consider a triangular array with observations that are i.i.d. within each row. The distribution of (x_i, z_i, w_i), the coefficient β, and the dimensions p, q, m are fixed; only the conditional distribution of ς_i may depend on n. Thus, the population projections, J, and K_M are fixed. The design distribution is held fixed, rather than the realized sample design.

Write $\mathbb{E}_n$ and Var_n for expectation and variance under the row-n distribution. Following Armstrong and Kolesár (2021), impose

$$\mathbb{E}_n(h_i \varsigma_i) = \frac{\tau}{\sqrt{n}}, \qquad \tau = (\tau_z^\top, \tau_w^\top)^\top \in \mathbb{R}^{2p}. \tag{3.6}$$

The vector τ is fixed and must be attainable under the maintained design distribution. For example, if $a^\top h_i = 0$ almost surely, then necessarily $a^\top \tau = 0$.

In this regime, the score definitions in (3.5) use $\mathbb{E}_n$ for the disturbance cross moments, while their projection operators remain fixed. Projection orthogonality gives $\mathbb{E}_n(\psi_i) = \tau/\sqrt{n}$. Thus estimating the projections changes the first-order covariance but not the local moment drift.

Assumption 4 (Combined GMM conditions under local misspecification)**.** Under the fixed population design distribution described above, suppose the following conditions hold.

(i) **Local restriction and identification.** Equation (3.6) holds for a fixed, attainable τ. The design has finite second moments, $\Sigma_z \in \mathbb{S}^q_{++}$, and $\Sigma_w \in \mathbb{S}^m_{++}$. The fixed symmetric weight $M \succeq 0$ satisfies $J^\top M J \in \mathbb{S}^p_{++}$.

(ii) **Moment conditions.** Suppose $\mathbb{E}\|x_i x_i^\top\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|z_i z_i^\top\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|w_i w_i^\top\|_{\mathrm{op}}^2 < \infty$, $\mathbb{E}\|x_i z_i^\top\|_{\mathrm{op}}^2 < \infty$, and $\mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty$. Fur-

ther, suppose $\sup_n \mathbb{E}_n(\varsigma_i^2) < \infty$, $\sup_n \mathbb{E}_n\|x_i\varsigma_i\|_2^2 < \infty$, $\sup_n \mathbb{E}_n\|z_i\{\varsigma_i - z_i^\top \Sigma_z^{-1}\mathbb{E}_n(z_i\varsigma_i)\}\|_2^2 < \infty$, and $\sup_n \mathbb{E}_n\|w_i\{\varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}_n(w_i\varsigma_i)\}\|_2^2 < \infty$.

(iii) **Score tails, covariance convergence, and nondegeneracy.** For some fixed $\eta > 0$, the scores satisfy $\sup_n \mathbb{E}_n\|\psi_i\|_2^{2+\eta} < \infty$. The score covariance satisfies $\mathrm{Var}_n(\psi_i) \to \Sigma_{\mathrm{miss}} \in \mathbb{S}_+^{2p}$, and $V_{\mathrm{miss}}(M) := K_M \Sigma_{\mathrm{miss}} K_M^\top \in \mathbb{S}_{++}^p$.

The moment conditions support sample moment convergence and the projection expansions, while the uniform $2+\eta$ score moment bound ensures the Lindeberg condition.

Theorem 5 (Combined GMM under local misspecification)**.** Under Assumption 4 (i), the following statements hold. (i) The population GMM criterion has the unique minimizer $\beta_{M,n} = \beta + n^{-1/2}K_M\tau$, and $\beta_{M,n} \to \beta$. (ii) If Assumption 4 (ii) also holds, then $\widehat{\beta}(M) \xrightarrow{\mathrm{p}} \beta$. (iii) Under the full assumption, $\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\mathrm{d}} \mathcal{N}\{K_M\tau, V_{\mathrm{miss}}(M)\}$.

The corresponding single-source conclusions hold under the same conditions when the selected block identifies the full coefficient.

Under exact specification, weighting affects first-order variance. Under local misspecification, it also determines the local bias $K_M\tau$. The two regimes share the same form of asymptotic linear representation, although

their covariance limits need not coincide. Setting $\tau = 0$ and $\Sigma_{\text{miss}} = \Sigma$ recovers the exact-specification limit.

4. Optimal weighting across specification regimes

We choose the weight using squared Euclidean loss. Since first-order behavior depends on M through K_M, the optimization can be expressed directly in terms of the sensitivity.

4.1 Risk-optimal weighting

For a fixed local sequence indexed by τ, define

$$\text{AMSE}_\tau(M) := \lim_{n\to\infty} n\,\mathbb{E}_\tau\|\widehat{\beta}(M) - \beta\|_2^2, \tag{4.1}$$

whenever the limit exists. Here $\mathbb{E}_\tau$ denotes expectation along the specified local sequence. If the squared root-n estimation errors are uniformly integrable, Theorem 5 gives

$$\text{AMSE}_\tau(M) = \|K_M\tau\|_2^2 + \text{tr}(K_M\Sigma_{\text{miss}}K_M^\top).$$

Let $D := \text{diag}(c_z I_p, c_w I_p)$, with $c_z, c_w \geq 0$, and $\mathcal{C}(D) := \{Du : \|u\|_2 \leq 1\}$. The radii are prespecified sensitivity parameters on the root-n projected-moment scale. When both radii are positive, the set imposes the joint budget $\|\tau_z\|_2^2/c_z^2 + \|\tau_w\|_2^2/c_w^2 \leq 1$. A zero radius sets the corresponding block's

violation to zero. The radii must be specified relative to the units of the moments.

For comparison across local directions, consider sequences sharing the same fixed design distribution and the same limiting score covariance Σ_{miss}. The latter is an additional restriction on the sequence family; it does not follow solely from fixing the design distribution.

Lemma 6 (Worst-case asymptotic risk)**.** For $K \in \mathbb{R}^{p\times 2p}$ satisfying $KJ = I_p$, recall its first-order risk by $\text{AMSE}_\tau(K) := \|K\tau\|_2^2 + \text{tr}(K\Sigma_{\text{miss}}K^\top)$. Then

$$R(K) := \sup_{\tau\in\mathcal{C}(D)} \text{AMSE}_\tau(K) = \|KD\|_{\text{op}}^2 + \text{tr}(K\Sigma_{\text{miss}}K^\top). \tag{4.2}$$

For an estimator with the stated local asymptotic distribution and sensitivity K, this first-order risk equals the limit in (4.1) when its squared root-n errors are uniformly integrable.

This criterion takes the supremum of pointwise asymptotic risks; it does not interchange a supremum and a finite-sample risk limit. If $\mathcal{C}(D)$ contains unattainable directions, $R(K)$ is a conservative envelope for the worst-case risk over the attainable directions.

The risk-optimal sensitivity solves

$$K^* \in \operatorname*{argmin}_{K:\, KJ=I_p} \left\{ \|KD\|_{\text{op}}^2 + \text{tr}(K\Sigma_{\text{miss}}K^\top) \right\}. \tag{4.3}$$

The constraint is affine and the objective is convex. If $\Sigma_{\text{miss}} \in \mathbb{S}^{2p}_{++}$, the objective is strongly convex and coercive, giving a unique minimizer whenever $\operatorname{rank}(J) = p$.

The positive-definite covariance condition used for uniqueness and feasible optimal weighting is stronger than the conditions required for complementary identification and fixed-weight inference. In particular, if Q_{sv} is singular, some nonzero $a \in \ker(Q_{\text{sv}})$ satisfies $a^\top \Pi_w^\perp x_i = 0$ almost surely, and hence $a^\top \psi_{\text{sv},i} = 0$ along the local sequence. Thus Σ_{miss} is singular, even though $\operatorname{rank}(J) = p$ may still hold. Such cases remain covered by the preceding identification and fixed-weight asymptotic results.

The optimization is equivalent to optimizing over positive-definite GMM weights. Every GMM sensitivity satisfies $K_M J = I_p$. Let $N_J \in \mathbb{R}^{2p \times p}$ have orthonormal columns spanning $\ker(J^\top)$. For any $KJ = I_p$, the positive-definite weights inducing K are exactly

$$\mathcal{M}(K) := \{K^\top A K + N_J B N_J^\top : A, B \in \mathbb{S}^p_{++}\}.$$

Each member is positive definite because $\ker(K) \cap \mathcal{C}(J) = \{0_{2p}\}$. Moreover, $J^\top M J = A$ and $J^\top M = AK$, so $(J^\top M J)^{-1} J^\top M = K$.

Conversely, for any positive-definite weight M inducing K, set $A := J^\top M J$. Then $A \in \mathbb{S}^p_{++}$, $J^\top M = AK$, and $M - K^\top A K = M - MJ(J^\top M J)^{-1} J^\top M$. Factoring out $M^{1/2}$ shows that this remainder is positive semidefinite with

null space $\mathcal{C}(J)$. It therefore equals $N_J B N_J^\top$ for some $B \in \mathbb{S}^p_{++}$, establishing the representation. The weights inducing the unique optimum are consequently $\mathcal{M}(K^*)$.

Under exact specification, take $D = 0$ and use the exact covariance Σ. When $\Sigma \in \mathbb{S}^{2p}_{++}$, the unique variance-optimal sensitivity is $K^* = (J^\top \Sigma^{-1} J)^{-1} J^\top \Sigma^{-1}$. The efficient-GMM weight $M^\star = \Sigma^{-1}$ is a canonical inducing weight (Hansen, 1982). Under local misspecification, the term $\|KD\|^2_{\text{op}}$ trades sensitivity to moment violations against precision (Armstrong and Kolesár, 2021).

When $Q_{\text{IV}} \in \mathbb{S}^p_{++}$, the IV-only sensitivity is $K^{(\text{IV})} = (Q_{\text{IV}}^{-1}, 0_{p\times p})$, with risk

$$R\{K^{(\text{IV})}\} = c_z^2 \|Q_{\text{IV}}^{-1}\|^2_{\text{op}} + \text{tr}(Q_{\text{IV}}^{-1} \Sigma_{\text{IV,miss}} Q_{\text{IV}}^{-1}).$$

Similarly, when $Q_{\text{SV}} \in \mathbb{S}^p_{++}$, the SV-only sensitivity is $K^{(\text{SV})} = (0_{p\times p}, Q_{\text{SV}}^{-1})$, with risk

$$R\{K^{(\text{SV})}\} = c_w^2 \|Q_{\text{SV}}^{-1}\|^2_{\text{op}} + \text{tr}(Q_{\text{SV}}^{-1} \Sigma_{\text{SV,miss}} Q_{\text{SV}}^{-1}).$$

The covariance blocks are the corresponding diagonal blocks of Σ_{miss}.

4.2 Practical optimal estimation

The risk-optimal rule can be implemented by replacing the population Jacobian and score covariance with consistent sample analogues. Choose a

fixed first-step weight $M_0 \in \mathbb{S}^{2p}_{++}$, for example $M_0 = I_{2p}$, and compute the preliminary estimator $\widehat{\beta}(M_0)$. Using the residuals $y_i - x_i^\top \widehat{\beta}(M_0)$, construct the plug-in versions $\widehat{\psi}_{\mathrm{IV},i}$ and $\widehat{\psi}_{\mathrm{SV},i}$ of the first-order scores in (3.5) by replacing the population projection matrices and disturbance cross moments with their sample analogues. Write $\widehat{\psi}_i := (\widehat{\psi}_{\mathrm{IV},i}^\top, \widehat{\psi}_{\mathrm{SV},i}^\top)^\top$, and estimate the limiting score covariance by

$$\widehat{\Sigma} := \frac{1}{n}\sum_{i=1}^{n}(\widehat{\psi}_i - \overline{\widehat{\psi}})(\widehat{\psi}_i - \overline{\widehat{\psi}})^\top, \qquad \overline{\widehat{\psi}} := \frac{1}{n}\sum_{i=1}^{n}\widehat{\psi}_i.$$

Retaining the prespecified D, define the feasible risk-optimal sensitivity by

$$\widehat{K}^* \in \operatorname*{argmin}_{K:\, K\widehat{J}=I_p} \left\{ \|KD\|_{\mathrm{op}}^2 + \operatorname{tr}(K\widehat{\Sigma}K^\top) \right\}.$$

The corresponding estimator is

$$\widehat{\beta}_{\mathrm{opt}} = \widehat{K}^* \frac{1}{n} \begin{pmatrix} X^\top P_Z Y \\ X^\top P_W^\perp Y \end{pmatrix}.$$

Thus the oracle bias–variance tradeoff is implemented by estimating the Jacobian and first-order score covariance and solving the sample analogue of (4.3).

Equivalently, on the event that $\operatorname{rank}(\widehat{J}) = p$, let $\widehat{N}_J \in \mathbb{R}^{2p\times p}$ have orthonormal columns spanning $\ker(\widehat{J}^\top)$. A canonical positive-definite GMM weight inducing $\widehat{K}^*$ is $\widehat{M}_{\mathrm{opt}} := \widehat{K}^{*\top}\widehat{K}^* + \widehat{N}_J\widehat{N}_J^\top$. Indeed, $(\widehat{J}^\top\widehat{M}_{\mathrm{opt}}\widehat{J})^{-1}\widehat{J}^\top\widehat{M}_{\mathrm{opt}} =$

$\widehat{K}^*$, so that $\widehat{\beta}(\widehat{M}_{\text{opt}}) = \widehat{\beta}_{\text{opt}}$. More generally, any sample analogue of $\mathcal{M}(\widehat{K}^*)$ yields the same estimator.

Proposition 7 (Feasible risk-optimal weighting)**.** Suppose Assumption 4 holds, $\Sigma_{\text{miss}} \in \mathbb{S}^{2p}_{++}$, and D is fixed. Let K^* denote the unique solution of (4.3). Then $\widehat{J} \xrightarrow{\text{p}} J$, $\widehat{\Sigma} \xrightarrow{\text{p}} \Sigma_{\text{miss}}$, and $\widehat{K}^* \xrightarrow{\text{p}} K^*$. Moreover, $\sqrt{n}\{\widehat{\beta}_{\text{opt}} - \beta\} \xrightarrow{\text{d}} \mathcal{N}\left(K^*\tau,\, K^*\Sigma_{\text{miss}}K^{*\top}\right)$. Hence estimating the risk-optimal sensitivity does not alter its first-order oracle limit.

Under exact specification, if $D = 0$ and $\Sigma_{\text{miss}} = \Sigma$, then $K^* = (J^\top\Sigma^{-1}J)^{-1}J^\top\Sigma^{-1}$, and therefore $\sqrt{n}\{\widehat{\beta}_{\text{opt}} - \beta\} \xrightarrow{\text{d}} \mathcal{N}\left(0,\, (J^\top\Sigma^{-1}J)^{-1}\right)$. Thus the feasible procedure reduces to efficient projected GMM when the sensitivity budget is zero.

4.3 Risk dominance across specification regimes

The single-source sensitivities are feasible for the combined optimization whenever their respective blocks identify the full coefficient.

Proposition 8 (Variance dominance under exact specification)**.** Suppose Assumption 3 holds for $M^\star = \Sigma^{-1}$, $Q_{\text{IV}}, Q_{\text{SV}} \in \mathbb{S}^p_{++}$, and $\Sigma \in \mathbb{S}^{2p}_{++}$. Then

$$\operatorname{tr}\{(J^\top\Sigma^{-1}J)^{-1}\} \le \min\left\{\operatorname{tr}(Q_{\text{IV}}^{-1}\Sigma_{\text{IV}}Q_{\text{IV}}^{-T}), \operatorname{tr}(Q_{\text{SV}}^{-1}\Sigma_{\text{SV}}Q_{\text{SV}}^{-T})\right\}. \tag{4.4}$$

Thus the efficient combined estimator has no larger trace of asymptotic

variance than either single-source estimator. Under uniform integrability of their squared root-n errors, the same comparison holds for AMSE.

Proposition 9 (Risk dominance under local misspecification)**.** Suppose $\operatorname{rank}(J) = p$, $Q_{\text{IV}}, Q_{\text{SV}} \in \mathbb{S}^p_{++}$, and $\Sigma_{\text{miss}} \in \mathbb{S}^{2p}_{++}$. For the same set $\mathcal{C}(D)$ and covariance Σ_{miss}, let K^* solve (4.3). Then $R(K^*) \leq R\{K^{(\text{IV})}\}$ and $R(K^*) \leq R\{K^{(\text{SV})}\}$.

When all directions in $\mathcal{C}(D)$ are attainable and the stated local asymptotic distributions and uniform-integrability conditions hold, these inequalities compare worst-case local AMSE. Otherwise, they compare the corresponding first-order risk envelopes.

Both results follow because the single-source sensitivities belong to $\{K : KJ = I_p\}$. The local result compares oracle worst-case criteria under the same budget; it does not assert dominance at every fixed τ or finite sample size. Convergence in distribution of the feasible estimator also does not, by itself, establish convergence of its mean squared error.

A comparison with one single-source procedure requires only that procedure's block to be nonsingular. The combined system can remain identified when individual blocks are singular, provided that $\operatorname{rank}(J) = p$.

Remark 1 (Confidence intervals allowing local misspecification)**.** For a prespecified nonzero $a \in \mathbb{R}^p$, the bias envelope is $\sup_{\tau \in \mathcal{C}(D)} |a^\top K\tau| = \|DK^\top a\|_2$.

For the feasible estimator, let $\widehat{b}_a := \|D\widehat{K}^\top a\|_2$ and $\widehat{s}_a^2 := a^\top \widehat{K}\widehat{\Sigma}\widehat{K}^\top a$.

Suppose $\widehat{K} \xrightarrow{\text{p}} K$, $\widehat{\Sigma} \xrightarrow{\text{p}} \Sigma_{\text{miss}}$, $a^\top K \Sigma_{\text{miss}} K^\top a > 0$, and the estimator has the stated local asymptotic normal distribution. Then the interval

$$a^\top \widehat{\beta} \pm \text{cv}_\alpha\left(\frac{\widehat{b}_a}{\widehat{s}_a}\right)\frac{\widehat{s}_a}{\sqrt{n}}$$

has asymptotic coverage at least $1-\alpha$ along each fixed attainable local sequence with $\tau \in \mathcal{C}(D)$. Here, $\text{cv}_\alpha(t)$ denotes the critical value of the folded normal distribution: for $Z \sim \mathcal{N}(t,1)$, $\text{cv}_\alpha(t)$ is the $(1-\alpha)$-quantile of $|Z|$, that is, $\Pr\{|Z| \le \text{cv}_\alpha(t)\} = 1-\alpha$. Equivalently, $\text{cv}_\alpha(t)$ is the solution c to $\Phi(c-t) - \Phi(-c-t) = 1-\alpha$, where Φ denotes the standard normal distribution function. In particular, $\text{cv}_\alpha(0) = z_{1-\alpha/2}$ (Armstrong and Kolesár, 2021). Uniform coverage additionally requires uniform asymptotic approximation and uniform consistency of the estimated ingredients. When $D = 0$, the interval reduces to the Wald interval.

4.4 Restricted weighting: a one-parameter WLS path

For interpretation, consider a weighted least-squares (WLS) path that combines the IV and SV projection criteria. We assume both population blocks are nonsingular so that the path includes both single-source endpoints. This full-path analysis excludes IV-rank-deficient settings such as $q < p$, although the stacked system may remain identified.

For $\lambda \in [0,1]$, define

$$\widehat{\beta}_\lambda := \operatorname*{argmin}_{\theta \in \mathbb{R}^p} \left\{ (1-\lambda)\|P_Z(Y - X\theta)\|_2^2 + \lambda \|P_W^\perp (Y - X\theta)\|_2^2 \right\}.$$

Whenever the inverse exists,

$$\widehat{\beta}_\lambda = \{(1-\lambda)X^\top P_Z X + \lambda X^\top P_W^\perp X\}^{-1}\{(1-\lambda)X^\top P_Z Y + \lambda X^\top P_W^\perp Y\}. \quad (4.5)$$

The endpoints satisfy $\widehat{\beta}_0 = \widehat{\beta}_{\text{IV}}$ and $\widehat{\beta}_1 = \widehat{\beta}_{\text{SV}}$.

Proposition 10 (Restricted-GMM interpretation)**.** Suppose $n^{-1}X^\top P_Z X \in \mathbb{S}^p_{++}$ and $n^{-1}X^\top P_W^\perp X \in \mathbb{S}^p_{++}$. For $\lambda \in [0,1]$, define

$$M_\lambda := \operatorname{diag}\left\{ (1-\lambda)(n^{-1}X^\top P_Z X)^{-1}, \lambda (n^{-1}X^\top P_W^\perp X)^{-1} \right\}.$$

Then $\widehat{\beta}_\lambda = \widehat{\beta}(M_\lambda)$. The weight is positive definite in the interior and positive semidefinite at the endpoints.

For fixed λ, sample moment consistency gives $M_\lambda \xrightarrow{\text{p}} \operatorname{diag}\{(1-\lambda)Q_{\text{IV}}^{-1}, \lambda Q_{\text{SV}}^{-1}\}$. Its limiting sensitivity is $\{(1-\lambda)Q_{\text{IV}} + \lambda Q_{\text{SV}}\}^{-1}\big((1-\lambda)I_p, \lambda I_p\big)$. Under the preceding regularity conditions, the random weight therefore has the same first-order behavior as its deterministic limit. A data-selected λ requires additional conditions on its convergence.

The WLS path restricts weighting to a block-diagonal one-parameter family. General GMM weights can also account for cross-block dependence through off-diagonal blocks.

For finite-sample conditional risk, let $S_Z := X^\top P_Z X$, $S_W := X^\top P_W^\perp X$, $S_\lambda := (1-\lambda)S_Z + \lambda S_W$, $\mathcal{D} := \sigma(X, Z, W)$, $U_Z := X^\top P_Z \varsigma$, and $U_W := X^\top P_W^\perp \varsigma$. With $U_\lambda := (1-\lambda)U_Z + \lambda U_W$,

$$\mathcal{R}_{\mathrm{tr}}(\lambda) := \mathbb{E}(\|\widehat{\beta}_\lambda - \beta\|_2^2 \mid \mathcal{D}) = \mathrm{tr}\{S_\lambda^{-1}\mathbb{E}(U_\lambda U_\lambda^\top \mid \mathcal{D})S_\lambda^{-1}\}. \tag{4.6}$$

The expression applies whenever the inverse and conditional second moments exist. In the local regime, conditional expectations are taken under the row-n distribution. This risk includes conditional bias: unconditional projected validity does not imply conditional centering.

Appendix E gives eigenvalue bounds and an oracle surrogate criterion for this conditional risk. That criterion describes the restricted path rather than the unrestricted GMM optimum.

5. Simulation

We consider two scalar designs illustrating the finite-sample implications of the weighting results. The first studies efficiency under correct specification, while the second introduces local misspecification in both the IV and SV moments and examines the resulting bias–variance tradeoff. OLS is included as a benchmark. All projections and data-dependent GMM weights are re-estimated in each of 1,000 Monte Carlo replications, with $n \in \{200, 600, 1000, 1400, 1800\}$.

5.1 Efficiency under correct specification

Consider

$$z_i = s_i + \eta_i, \qquad w_i = (u_i, s_i)^\top,$$

$$x_i = u_i + 2s_i + v_i, \qquad y_i = \beta x_i + u_i + \varepsilon_i.$$

where $s_i, \eta_i, u_i, v_i, \varepsilon_i$ are mutually independent standard normal variables. The IV-only and SV-only estimators both have asymptotic variance 1, compared with 0.68 for equal-weight GMM and 0.50 for efficient GMM. We report MSE at $\beta = 1$, and two-sided 5% Wald rejection probabilities under $H_0 : \beta = 0$ and the local alternative $\beta = 2/\sqrt{n}$.

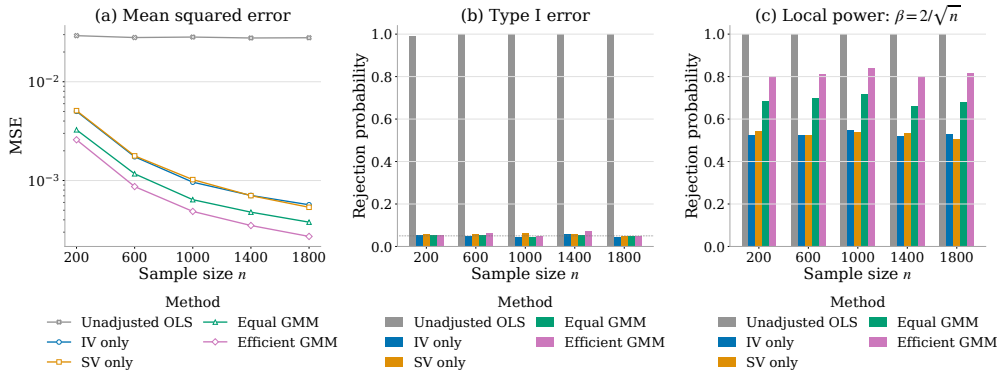


Figure 2: Performance under correct specification. Panel (a) reports MSE at $\beta = 1$ on a logarithmic scale; panels (b) and (c) report rejection probabilities under $H_0 : \beta = 0$ and $\beta = 2/\sqrt{n}$, respectively.

Figure 2 confirms the gains from combining the two moment blocks: efficient GMM attains the lowest MSE and highest local power among size-controlled procedures, while equal-weight GMM also improves on either single-source estimator. OLS remains biased because x_i contains u_i; its bias is $1/6$, producing persistent MSE and severe size distortion.

5.2 Risk under local misspecification

We next allow local misspecification of both the IV and SV moments. Following Lemma 6, let $D = \mathrm{diag}(1, 1)$, $\mathcal{C}(D) = \{(\tau_z, \tau_w)^\top : \tau_z^2 + \tau_w^2 \leq 1\}$, so both channels may deviate. We compare OLS, IV only, SV only, equal-weight GMM, and feasible local-risk GMM. At $\beta = 1$, we report worst-case MSE and Type I error for $H_0 : \beta = 1$, together with minimum local power under $\beta = 1 + 2/\sqrt{n}$, evaluated over 16 boundary directions. OLS uses conventional Wald inference, whereas the remaining methods use the misspecification-robust intervals in Remark 1.

Across the reported sample sizes, feasible local-risk GMM has the lowest estimated worst-case MSE among the methods compared (Figure 3). This pattern agrees with the oracle first-order risk comparison in Proposition 9. The robust procedures maintain approximately nominal size, with some finite-sample overrejection, whereas OLS remains distorted. Local misspec-

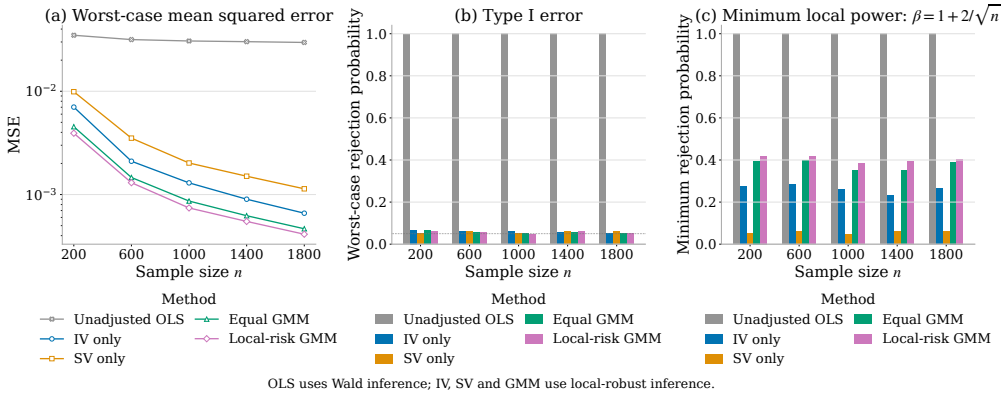


Figure 3: Performance under local misspecification of both the IV and SV moments. Panel (a) reports worst-case MSE on a logarithmic scale; panels (b) and (c) report worst-case Type I error and minimum local power over the evaluated boundary directions.

ification substantially reduces single-source power, especially for SV only, while the combined procedures retain more power. Local-risk GMM has higher minimum local power than equal-weight GMM here, although the former optimizes worst-case estimation risk rather than testing power.

6. Real data

We apply the proposed methods to the K562 essential-scale CRISPRi Perturb-seq experiment of Replogle et al. (2022). In this experiment, paired single-guide RNAs repress individual target genes, and single-cell RNA sequencing with direct guide capture jointly measures gene expression and perturbation identity six days after infection. We analyze pseudobulk expression profiles constructed within perturbation–batch groups across 48 sequencing batches. After filtering, the analysis includes 213 perturbation targets and a common panel of 2,000 candidate downstream genes. Each target-specific analysis combines 30–48 perturbed profiles with 48 control profiles, giving $n = 78$–96. Details of pseudobulk construction, gene selection, and preprocessing are provided in Appendix F.

For each perturbation–outcome pair, we fit

$$y_i = \beta x_i + \varsigma_i,$$

where x_i and y_i denote target and downstream gene expression, respectively,

and z_i indicates perturbation versus control. The surrogate controls w_i are the first 10 principal component scores of a common panel of 1,000 genes selected for low differential-expression responses across perturbations. The perturbation indicator provides the IV channel, whereas residualization on w_i provides the SV channel. We compare the IV, SV, and optimally weighted two-step GMM estimators over $213 \times 2,000 = 426,000$ pairs, with standard errors clustered by batch.

To assess sensitivity to the maintained moment restrictions, we supplement Wald inference with local misspecification-adjusted inference under

$$\sqrt{n}\,\mathbb{E}\{g_i(\beta_0)\} = \tau, \qquad \tau = \mathrm{diag}(r_{\mathrm{IV}}, r_{\mathrm{SV}})u, \qquad \|u\|_2 \leq 1,$$

where $r_j = 0.5\sqrt{\widehat{\Omega}_{jj}}$ and $\widehat{\Omega}$ is the batch-clustered covariance of the stacked moment scores evaluated at the GMM estimate. The same pair-specific deviation set is used for all three estimators. For each perturbation, method, and inference rule, we apply the Benjamini–Hochberg procedure across the 2,000 outcomes at a nominal false discovery rate of 5%.

Figure 4 shows substantial overlap among the three methods, together with method-specific discoveries. Wald GMM identifies 84,363 significant pairs, including some not selected by either single-source estimator. Local misspecification-adjusted inference retains 70,132 GMM discoveries, a reduction of approximately 16.9%. Thus, most GMM discoveries persist

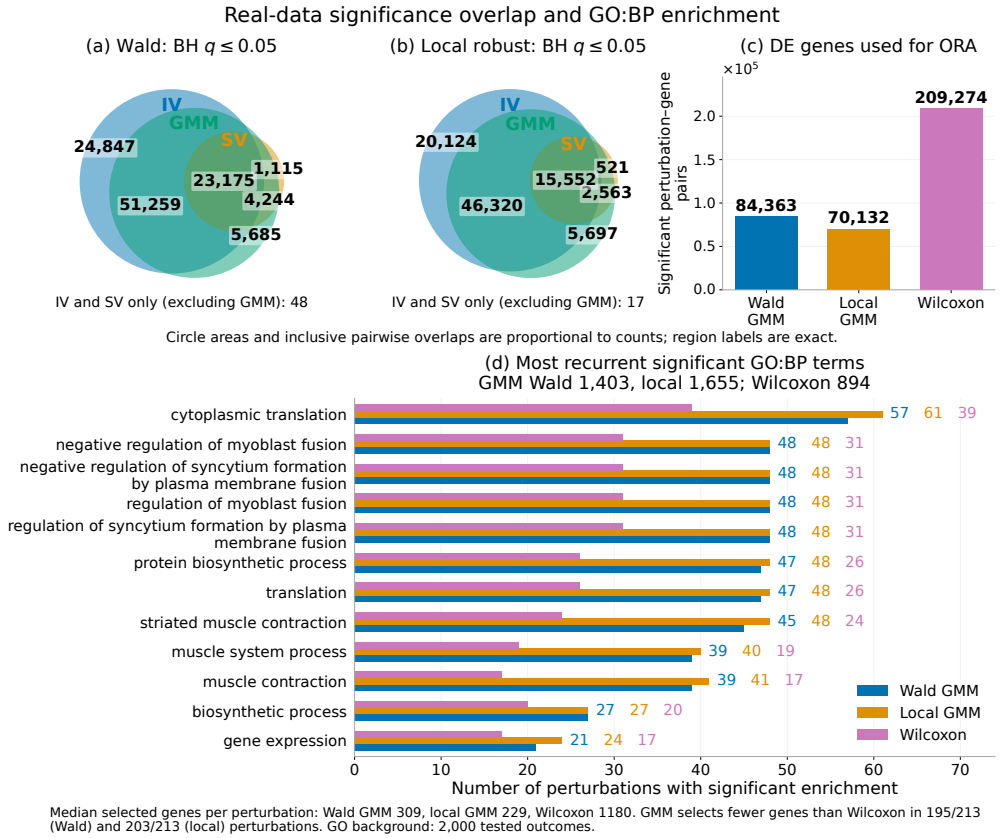


Figure 4: Overlap of significant perturbation–outcome pairs and Gene Ontology biological-process enrichment. Panels (a)–(b) compare IV, SV, and GMM discoveries under Wald and local misspecification-adjusted inference, respectively; panel (c) compares GMM with batch-stratified Wilcoxon differential expression on the same outcome panel; panel (d) summarizes recurrent significant GO biological-process terms.

under the chosen calibration of local moment deviations.

The batch-stratified Wilcoxon analysis identifies 209,274 significant pairs on the same testing panel. This comparison concerns different statistical targets: Wilcoxon detects perturbation-associated expression differences, whereas GMM tests the target-expression coefficient under the maintained moment restrictions. Despite its smaller discovery set, GMM yields recurrent GO biological-process enrichment, particularly for translation- and biosynthesis-related processes. These main biological themes remain similar after adjustment for local misspecification.

7. Discussion

This paper provides a common projected-moment framework for deconfounding with instruments and surrogate controls. Their distinct projection geometries allow complementary identification, while joint weighting can improve efficiency under correct specification and balance bias against variance under local misspecification. These benefits depend on the validity of the moment restrictions and the information supplied by the ancillary variables. In particular, the local sensitivity analysis quantifies departures within a prespecified set; it does not establish the validity of either source from the data.

An important extension concerns nonlinear structural models. A linear approximation may provide a useful summary over a restricted range of exposure values, but the resulting coefficient need not represent a constant causal effect. Moreover, instruments and surrogate controls may select different features of a nonlinear response, so their single-source targets need not coincide. Extending the joint framework would therefore require specifying a common target and deriving appropriate moment restrictions before combining them. Basis expansions offer one possible approach, although approximation error and the validity of the expanded moment conditions would require additional analysis.

Measurement error provides another direction for future work, particularly when exposures and surrogate controls are constructed from noisy biological measurements. Such errors can alter both the projection geometry and the validity of the resulting moments. The corrected-moment GMM approach of Evdokimov and Zeleneev (2025) suggests a possible route to extending the present framework. This would require assumptions on the measurement error mechanism, corrections to the affected moments, and an asymptotic analysis accounting for estimation of the correction terms. Studying how these corrections interact with complementary identification and risk-optimal weighting remains an open problem.

Supplementary Material

The supplementary material contains additional discussion of related work, proofs of all theoretical results, and further details on simulation and real-data analysis.

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A. Additional discussion on single-source deconfounding and causal interpretation

A.1 Additional details on the projected IV formulation

This subsection provides additional discussion of the projected IV formulation used in Section 2.2, including its relation to the classical IV moment, an explicit first-stage representation, and the simplification of the first-order theory under valid instruments.

Dimension and identification of the projected IV moment. Recall that $Q_{\text{IV}} = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$. When $\Sigma_z \in \mathbb{S}^q_{++}$, $\operatorname{rank}(Q_{\text{IV}}) = \operatorname{rank}(\Sigma_{zx}) \le \min(p, q)$. Hence, identification of the full p-dimensional coefficient vector from the IV block alone requires $\operatorname{rank}(Q_{\text{IV}}) = p$, and therefore necessarily $q \ge p$.

When $q < p$, the projected IV moment $\mathbb{E}[(\Pi_z x_i)(y_i - x_i^\top \theta)]$ is still written as a p-dimensional vector, but its Jacobian has rank at most $q < p$. Thus, the p displayed equations contain fewer than p independent identifying directions and cannot uniquely identify the full coefficient vector.

The distinction between the raw and projected IV formulations is also relevant when $q \ge p$. The classical IV restriction $\mathbb{E}[z_i(y_i - x_i^\top \theta)] = 0_q$ contains q moment equations and, under $\operatorname{rank}(\Sigma_{zx}) = p$, is just identified when $q = p$ and overidentified when $q > p$. By contrast, $\mathbb{E}[(\Pi_z x_i)(y_i - x_i^\top \theta)] = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}[z_i(y_i - x_i^\top \theta)]$ is always p-dimensional. Under valid IV, $\mathbb{E}(z_i \varsigma_i) = 0_q$, the raw and projected moments identify the same structural coefficient under the corresponding rank condition. When $q > p$, however, projection may discard overidentifying di-

rections contained in the raw IV moment. Consequently, efficiency statements for the projected IV system are not, in general, efficiency statements for GMM based on all raw IV moments.

Relation to an explicit first-stage representation. Some formulations of instrumental variables additionally posit $x_i = \eta^\top z_i + \epsilon_i$ with $\mathbb{E}(z_i \epsilon_i^\top) = 0$. Such a representation is useful when a structural interpretation of the first-stage regression is desired, but it is not required for the projected-moment formulation used here.

Indeed, whenever $\Sigma_z \in \mathbb{S}^q_{++}$, the population linear projection is defined directly as $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$. Thus, the projected-IV statistical target and the corresponding 2SLS estimating equation depend only on this projection, the projected moment bias $\delta_{\text{IV}} := \mathbb{E}[(\Pi_z x_i)\varsigma_i]$, and the relevance matrix $Q_{\text{IV}} := \mathbb{E}[(\Pi_z x_i)x_i^\top]$. Accordingly, an explicit decomposition $x_i = \eta^\top z_i + \epsilon_i$ may be viewed as an optional linear-projection representation rather than a primitive assumption.

First-order representation under valid IV. Under the general projected-IV target, the first-order term is $\psi_{\text{IV},i} = (\Pi_z x_i)e_{\text{IV},i} + x_i z_i^\top \Sigma_z^{-1}\mathbb{E}[z_i(y_i - x_i^\top \beta_{\text{IV}})]$. The condition $\mathbb{E}[\|x_i z_i^\top\|_{\text{op}}^2] < \infty$ controls the second moment of the second component, whereas $\mathbb{E}\|z_i e_{\text{IV},i}\|_2^2 < \infty$ controls the projected-score component because $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$.

Under valid IV, $\mathbb{E}(z_i\varsigma_i) = 0_q$, we have $\delta_{\text{IV}} = 0_p$, $\beta_{\text{IV}} = \beta$, and $\mathbb{E}[z_i(y_i - x_i^\top \beta_{\text{IV}})] = 0_q$. Hence, the correction term vanishes and $e_{\text{IV},i} = \varsigma_i$, so that $\psi_{\text{IV},i} = (\Pi_z x_i)\varsigma_i$. Therefore, $\mathbb{E}[\|x_i z_i^\top\|_{\text{op}}^2] < \infty$ is unnecessary for asymptotic normality in the valid-IV case; it suffices to impose the corresponding finite-second-moment condition on $z_i\varsigma_i$, together with nondegeneracy of the limiting

covariance matrix.

A.2 Additional details on surrogate-control residualization

This subsection provides further interpretation of the surrogate-control formulation in Section 2.3, with emphasis on the geometry of residualization, its projection representation, and the simplification of the first-order theory under exact surrogate-control validity.

Residualized relevance and projection geometry. Recall that $\Pi_w^{\perp} x_i = x_i - \Sigma_{xw}\Sigma_w^{-1} w_i$ and $Q_{\text{SV}} = \mathbb{E}[(\Pi_w^{\perp} x_i) x_i^{\top}]$. Since the population residual $\Pi_w^{\perp} x_i$ is orthogonal to w_i, we have

$$Q_{\text{SV}} = \mathbb{E}[(\Pi_w^{\perp} x_i)(\Pi_w^{\perp} x_i)^{\top}].$$

Hence, $Q_{\text{SV}} \in \mathbb{S}_{++}^{p}$ requires the residualized regressor to retain variation in every direction of the p-dimensional coefficient space.

This differs geometrically from IV relevance. The IV projection retains the component of x_i explained by z_i, whereas surrogate-control residualization removes the component of x_i explained by w_i. Thus, IV identification requires the projected component to span enough directions, whereas SV identification requires sufficiently rich variation to remain after partialling out the surrogate controls. This distinction motivates combining the two sources in the stacked moment system developed in the main text.

Partialling out as a projection representation. The surrogate-control estimator does not require a structural equation describing how w_i generates x_i. Its population construction only requires the linear projection $\Pi_w x_i = \Sigma_{xw}\Sigma_w^{-1} w_i$ and the residual $\Pi_w^{\perp} x_i = x_i - \Pi_w x_i$.

Equivalently, writing $\Gamma_w := \Sigma_w^{-1}\Sigma_{wx}$, one has the population decomposition $x_i = \Gamma_w^{\top} w_i + \Pi_w^{\perp} x_i$, with $\mathbb{E}[(\Pi_w^{\perp} x_i) w_i^{\top}] = 0$. The statistical target is determined by the residualized moment $\mathbb{E}[(\Pi_w^{\perp} x_i)(y_i - x_i^{\top}\theta)]$, rather than by a structural model for the relationship between x_i and w_i.

In particular, w_i itself need not be orthogonal to the structural disturbance. Under exact surrogate-control validity, $\mathbb{E}[(\Pi_w^{\perp} x_i)\varsigma_i] = 0_p$, which is equivalent to

$$\mathbb{E}(x_i \varsigma_i) = \Sigma_{xw}\Sigma_w^{-1}\mathbb{E}(w_i \varsigma_i).$$

Thus, the surrogate controls absorb the component of regressor–disturbance dependence represented linearly through w_i, leaving residualized variation orthogonal to the disturbance.

First-order representation under exact surrogate-control validity. Under the general statistical target $\beta_{\mathrm{SV}} = \beta + Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}$, define $r_{\mathrm{SV},i} := y_i - x_i^{\top}\beta_{\mathrm{SV}}$ and

$$e_{\mathrm{SV},i} := r_{\mathrm{SV},i} - w_i^{\top}\Sigma_w^{-1}\mathbb{E}(w_i r_{\mathrm{SV},i}).$$

Then $\mathbb{E}(w_i e_{\mathrm{SV},i}) = 0_m$, and the first-order term is $\psi_{\mathrm{SV},i} = (\Pi_w^{\perp} x_i) e_{\mathrm{SV},i}$.

The simplification relative to the IV case follows from the exact annihilation property $P_W^{\perp} W = 0$. After recentering $r_{\mathrm{SV},i}$ with respect to w_i, the component in the surrogate-control

span is removed by the sample residual-maker. By contrast, $P_Z Z = Z$, so the corresponding instrument-span component is retained in the IV case and generates an additional first-order correction term under projected moment bias.

Under exact surrogate-control validity, $\delta_{\text{SV}} = 0_p$, we have $\beta_{\text{SV}} = \beta$ and $r_{\text{SV},i} = \varsigma_i$, so

$$e_{\text{SV},i} = \varsigma_i - w_i^\top \Sigma_w^{-1} \mathbb{E}(w_i \varsigma_i).$$

Exact SV validity does not generally imply $\mathbb{E}(w_i \varsigma_i) = 0_m$, and hence $e_{\text{SV},i}$ need not equal ς_i. The first-order term therefore remains $\psi_{\text{SV},i} = (\Pi_w^\perp x_i) e_{\text{SV},i}$.

However, because $r_{\text{SV},i} = \varsigma_i$, the condition $\mathbb{E}\|x_i x_i^\top\|_{\text{op}}^2 < \infty$, which is needed under the general target to control the contribution of $x_i^\top Q_{\text{SV}}^{-1} \delta_{\text{SV}}$, is unnecessary for asymptotic normality under exact surrogate-control validity. The remaining second-moment conditions continue to control $x_i \varsigma_i$, the interaction between x_i and w_i, and the residualized score.

A.3 Additional discussion on causal interpretation

The results in Sections 2.2 and 2.3 are stated in terms of statistical moment conditions and statistical targets. This subsection provides additional discussion of the structural assumptions under which the no-bias conditions $\delta_{\text{IV}} = 0$ and $\delta_{\text{SV}} = 0$ admit causal interpretations.

IV moments and causal direction. In the linear IV formulation, exogeneity is commonly represented by $\mathbb{E}(z_i \varsigma_i) = 0_q$, while the projected formulation used in this paper requires the corresponding projected no-bias condition $\delta_{\text{IV}} := \mathbb{E}[(\Pi_z x_i) \varsigma_i] = 0_p$. These moment

restrictions are statistical statements and do not, by themselves, encode a causal direction between z_i and x_i.

The causal structure matters because it determines whether such orthogonality is scientifically plausible. For example, the classical IV interpretation views z_i as generating variation in x_i while remaining unrelated to the latent determinants of the outcome. By contrast, if z_i is constructed from, or strongly driven by, the endogenous regressor in a data-generating mechanism resembling $\varsigma_i \to x_i \to z_i$, then z_i may inherit information about ς_i, making the no-bias restriction difficult to justify. In such cases, the projected estimator remains statistically defined, but its target is $\beta_{\text{IV}} = \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}}$.

Heterogeneous causal effects under IV. A common structural interpretation of β additionally relies on a homogeneous linear response. If the potential outcome instead takes the form $y_i(x) = \alpha_i + \beta_i^\top x$, or is nonlinear in x, the IV coefficient generally need not represent a universal structural slope.

For binary treatment, under the usual additional conditions including monotonicity, IV identifies a local average treatment effect for compliers rather than a population-wide common effect (Imbens and Angrist, 1994). More generally, linear IV should be interpreted as summarizing causal responses associated with the variation in x_i induced by the instrument, rather than automatically as the entire causal response surface.

This distinction is separate from moment invalidity. Even when $\delta_{\text{IV}} = 0$, effect heterogeneity may alter the causal interpretation of the IV estimand; when $\delta_{\text{IV}} \neq 0$, the target additionally

contains the shift $Q_{\text{IV}}^{-1}\delta_{\text{IV}}$ induced by violation of the projected moment restriction.

Surrogate controls and causal direction. For surrogate-control adjustment, the statistical no-bias condition is $\delta_{\text{SV}} := \mathbb{E}[(\Pi_w^{\perp} x_i)\varsigma_i] = 0_p$. A natural causal justification is that w_i is a pre-treatment proxy for the latent confounding structure and captures the component of x_i associated with the structural disturbance.

The temporal or causal role of w_i is therefore important. If the data-generating mechanism is instead closer to $\varsigma_i \to x_i \to w_i$, then w_i may contain variation generated by the endogenous regressor itself. Partialling out such a variable need not remove only latent confounding and may invalidate the desired causal interpretation. The estimator nevertheless remains centered at the statistical target $\beta_{\text{SV}} = \beta + Q_{\text{SV}}^{-1}\delta_{\text{SV}}$, with $Q_{\text{SV}}^{-1}\delta_{\text{SV}}$ representing the remaining departure from the structural coefficient.

Relation to auxiliary-variable and proximal approaches. The surrogate-control formulation considered here is related to auxiliary-variable and proxy-based approaches to latent confounding. In particular, the auxiliary-variable framework of Miao et al. (2023) accommodates variables satisfying suitable exclusion-type conditions and develops nonparametric identification through equivalence and completeness conditions. At a high level, both approaches use observed ancillary information without requiring direct recovery of the latent confounder.

The identification targets and mechanisms are nevertheless different. The framework of

A.3 Additional discussion on causal interpretation

Miao et al. (2023) concerns nonparametric identification of potential-outcome distributions, whereas the present paper focuses on a linear structural coefficient and the projection, rank, first-order, and weighting structure generated by IV and surrogate-control moments.

The present formulation is also related in spirit to proximal causal inference (Miao et al., 2018, 2024). Proximal methods typically introduce treatment-inducing and outcome-inducing proxies and identify causal effects through conditional moment equations involving a bridge function. Their identification theory relies on conditions such as proxy validity, existence of an appropriate bridge function, and completeness.

Our framework instead uses the direct residualized moment $\mathbb{E}[(\Pi_w^{\perp} x_i)(y_i - x_i^{\top}\theta)] = 0_p$. We therefore do not solve a nonparametric bridge equation or attempt to recover the latent confounder. The surrogate variable enters through linear projection geometry: the component of x_i explained by w_i is removed, and inference uses the remaining variation.

Accordingly, these approaches should be viewed as complementary rather than equivalent. Auxiliary-variable and proximal methods provide broader nonparametric causal identification theories under proxy-variable assumptions, whereas the present formulation imposes a linear working structure to obtain a transparent projected-moment representation that is directly comparable with the IV channel and can subsequently be combined with it within the unified GMM framework.

B. Notation

Population L_2 projections are denoted by Π_z, Π_w, and $\Pi_w^{\perp}$, whereas for any sample matrix A with n rows, $\mathcal{C}(A)$ denotes its column space, P_A the Euclidean projection onto $\mathcal{C}(A)$, and $P_A^{\perp} := I_n - P_A$ the corresponding residual-maker. For any matrix A, $\ker(A) := \{v : Av = 0\}$ denotes its null space and $\operatorname{rank}(A) := \dim \mathcal{C}(A)$ its rank.

We use $\|\cdot\|_2$, $\|\cdot\|_{\mathrm{F}}$, and $\|\cdot\|_{\mathrm{op}}$ for the Euclidean, Frobenius, and operator norms, respectively, and $\operatorname{Cov}(a, b)$ for population cross-covariance.

We write $\xrightarrow{\mathrm{p}}$ and $\xrightarrow{\mathrm{d}}$ for convergence in probability and distribution, use $O_p(\cdot)$, $o_p(\cdot)$, and $\Theta_p(\cdot)$ in their standard senses, and write $a_n \lesssim b_n$ and $a_n \asymp b_n$ for one-sided and two-sided order equivalence up to constants.

The sets of $d \times d$ positive definite and positive semidefinite matrices are denoted by $\mathbb{S}_{++}^d$ and $\mathbb{S}_{+}^d$, respectively.

We use lower-case letters for unit-level variables and upper-case letters for sample arrays: $y_i \in \mathbb{R}$, $x_i \in \mathbb{R}^p$, $z_i \in \mathbb{R}^q$, $w_i \in \mathbb{R}^m$, and $u_i \in \mathbb{R}^r$ denote the outcome, regressors, instruments, surrogate controls, and latent confounders, respectively, with Y, X, Z, W, U denoting their corresponding sample arrays.

C. Proof

C.1 Proof of Theorem 1

Proof of Theorem 1. Recall that $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$, $Q_{\text{IV}} = \mathbb{E}[(\Pi_z x_i)x_i^\top] = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$, $\delta_{\text{IV}} = \mathbb{E}[(\Pi_z x_i)\varsigma_i]$, and $\beta_{\text{IV}} = \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}}$.

(i) Identification. Recall that the projected IV population moment condition is $\mathbb{E}[(\Pi_z x_i)(y_i - x_i^\top\theta)] = 0_p$, where $\theta \in \mathbb{R}^p$ is a candidate coefficient vector. Using $y_i = x_i^\top\beta + \varsigma_i$, we have

$$\mathbb{E}\Big[(\Pi_z x_i)\big(y_i - x_i^\top\theta\big)\Big] = \mathbb{E}\Big[(\Pi_z x_i)x_i^\top\Big](\beta - \theta) + \mathbb{E}[(\Pi_z x_i)\varsigma_i]\,.$$

By the definitions $Q_{\text{IV}} := \mathbb{E}[(\Pi_z x_i)x_i^\top]$ and $\delta_{\text{IV}} := \mathbb{E}[(\Pi_z x_i)\varsigma_i]$, the population moment condition is therefore equivalent to

$$Q_{\text{IV}}(\theta - \beta) = \delta_{\text{IV}}. \tag{C.1}$$

By Assumption 1(i), $Q_{\text{IV}} \in \mathbb{S}^p_{++}$, and hence Q_{IV} is nonsingular. Left-multiplying (C.1) by Q_{IV}^{-1} gives

$$\theta = \beta + Q_{\text{IV}}^{-1}\delta_{\text{IV}} = \beta_{\text{IV}}.$$

Thus a solution exists.

To see that the solution is unique, suppose that both θ_1 and θ_2 satisfy (C.1). Then $Q_{\text{IV}}(\theta_1 - \theta_2) = 0_p$. Since Q_{IV} is nonsingular, its null space contains only 0_p, so $\theta_1 - \theta_2 = 0_p$, and hence $\theta_1 = \theta_2$. Therefore, the projected IV population moment uniquely identifies β_{IV}.

Finally, if $\delta_{\text{IV}} = 0_p$, then $Q_{\text{IV}}(\beta_{\text{IV}} - \beta) = 0_p$. Nonsingularity of Q_{IV} implies $\beta_{\text{IV}} = \beta$. Hence,

when the projected IV moment bias vanishes, the projected IV population moment identifies the structural coefficient β.

Define $r_{\text{IV},i} := y_i - x_i^\top \beta_{\text{IV}} = \varsigma_i - x_i^\top Q_{\text{IV}}^{-1}\delta_{\text{IV}}$, and write $r_{\text{IV}} := (r_{\text{IV},1}, \ldots, r_{\text{IV},n})^\top = Y - X\beta_{\text{IV}}$. Then

$$\mathbb{E}[(\Pi_z x_i) r_{\text{IV},i}] = \delta_{\text{IV}} - Q_{\text{IV}} Q_{\text{IV}}^{-1}\delta_{\text{IV}} = 0_p.$$

Hence, β_{IV} satisfies the population projected normal equation

$$\Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i}) = 0_p. \tag{C.2}$$

Importantly, (C.2) does not generally imply $\mathbb{E}(z_i r_{\text{IV},i}) = 0_q$.

Using $Y = X\beta_{\text{IV}} + r_{\text{IV}}$, the 2SLS estimator satisfies the exact decomposition

$$\widehat{\beta}_{\text{IV}} - \beta_{\text{IV}} = \left(\frac{1}{n}X^\top P_Z X\right)^{-1}\left(\frac{1}{n}X^\top P_Z r_{\text{IV}}\right). \tag{C.3}$$

(ii) Consistency. Suppose Assumption 1 (i) holds. We first consider the denominator in (C.3).

Under the maintained second-moment conditions, the Cauchy–Schwarz inequality gives $\mathbb{E}\|x_i z_i^\top\|_{\text{F}} \le \{\mathbb{E}\|x_i\|_2^2\}^{1/2}\{\mathbb{E}\|z_i\|_2^2\}^{1/2} < \infty$, while $\mathbb{E}\|z_i z_i^\top\|_{\text{F}} = \mathbb{E}\|z_i\|_2^2 < \infty$. Therefore, the weak law of large numbers applied entrywise gives $\frac{1}{n}X^\top Z \xrightarrow{\text{p}} \Sigma_{xz}$, $\frac{1}{n}Z^\top Z \xrightarrow{\text{p}} \Sigma_z$, and $\frac{1}{n}Z^\top X \xrightarrow{\text{p}} \Sigma_{zx}$. Since $\Sigma_z \in \mathbb{S}_{++}^q$, the continuous mapping theorem gives $(\frac{1}{n}Z^\top Z)^{-1} \xrightarrow{\text{p}} \Sigma_z^{-1}$. Thus,

$$\frac{1}{n}X^\top P_Z X = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\left(\frac{1}{n}Z^\top X\right) \xrightarrow{\text{p}} Q_{\text{IV}}. \tag{C.4}$$

Because $Q_{\text{IV}} \in \mathbb{S}_{++}^p$, $(\frac{1}{n}X^\top P_Z X)^{-1} \xrightarrow{\text{p}} Q_{\text{IV}}^{-1}$.

We next consider the numerator. Since $r_{\text{IV},i} = \varsigma_i - x_i^\top Q_{\text{IV}}^{-1}\delta_{\text{IV}}$, we have $\mathbb{E}(r_{\text{IV},i}^2) \le 2\mathbb{E}(\varsigma_i^2) + 2\|Q_{\text{IV}}^{-1}\delta_{\text{IV}}\|_2^2\mathbb{E}\|x_i\|_2^2 < \infty$. It follows from Cauchy–Schwarz that $\mathbb{E}\|z_i r_{\text{IV},i}\|_2 < \infty$, and hence $\frac{1}{n}\sum_{i=1}^n z_i r_{\text{IV},i} \xrightarrow{\text{p}} \mathbb{E}(z_i r_{\text{IV},i})$. Therefore,

$$\frac{1}{n}X^\top P_Z r_{\text{IV}} = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\frac{1}{n}\sum_{i=1}^n z_i r_{\text{IV},i} \xrightarrow{\text{p}} \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i}) = 0_p,$$

where the last equality follows from (C.2).

Combining this result with (C.4) and (C.3) gives $\widehat{\beta}_{\text{IV}} - \beta_{\text{IV}} \xrightarrow{\text{p}} 0_p$. Hence, $\widehat{\beta}_{\text{IV}} \xrightarrow{\text{p}} \beta_{\text{IV}}$.

(iii) Asymptotic normality. Suppose further that Assumption 1 (ii) holds. Multiplying (C.3) by $\sqrt{n}$ gives

$$\sqrt{n}(\widehat{\beta}_{\text{IV}} - \beta_{\text{IV}}) = \left(\frac{1}{n}X^\top P_Z X\right)^{-1}\left(\frac{1}{\sqrt{n}}X^\top P_Z r_{\text{IV}}\right). \tag{C.5}$$

By (C.4), the first factor converges in probability to Q_{IV}^{-1}. It therefore remains to derive the limiting distribution of the normalized numerator.

Directly expanding the numerator gives

$$\frac{1}{\sqrt{n}}X^\top P_Z r_{\text{IV}} = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n z_i r_{\text{IV},i}. \tag{C.6}$$

The projected normal equation (C.2) only requires $\Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i}) = 0_p$, while $\mathbb{E}(z_i r_{\text{IV},i})$ itself need not vanish. Consequently, $n^{-1/2}\sum_{i=1}^n z_i r_{\text{IV},i}$ may contain a mean component of order $\sqrt{n}$, so this sum is not generally $O_p(1)$. Thus, unlike a directly centered CLT term, the random projection coefficient in (C.6) cannot immediately be replaced by its probability limit.

To obtain a centered term, define

$$e_{\text{IV},i} := r_{\text{IV},i} - z_i^\top \Sigma_z^{-1} \mathbb{E}(z_i r_{\text{IV},i}).$$

The consistency argument gives $\mathbb{E}\|z_i r_{\text{IV},i}\|_2 < \infty$, so $\mathbb{E}(z_i r_{\text{IV},i})$ is finite. By construction,

$$\mathbb{E}(z_i e_{\text{IV},i}) = \mathbb{E}(z_i r_{\text{IV},i}) - \Sigma_z \Sigma_z^{-1} \mathbb{E}(z_i r_{\text{IV},i}) = 0_q.$$

Assumption 1 (ii) further gives $\mathbb{E}\|z_i e_{\text{IV},i}\|_2^2 < \infty$. Hence, the multivariate central limit theorem implies $n^{-1/2} \sum_{i=1}^n z_i e_{\text{IV},i} = O_p(1)$.

Writing $e_{\text{IV}} := (e_{\text{IV},1}, \ldots, e_{\text{IV},n})^\top$, the preceding definition gives $r_{\text{IV}} = e_{\text{IV}} + Z\Sigma_z^{-1}\mathbb{E}(z_i r_{\text{IV},i})$. At this point, the IV argument differs from the corresponding SV argument. For SV, the residual-maker eliminates the ancillary-variable component because $P_W^\perp W = 0$. For IV, however, $P_Z Z = Z$, so the component in the span of Z is retained. Substituting the decomposition of r_{IV} into (C.6) therefore gives the exact decomposition

$$\begin{aligned} \frac{1}{\sqrt{n}} X^\top P_Z r_{\text{IV}} = \left(\frac{1}{n} X^\top Z\right) \left(\frac{1}{n} Z^\top Z\right)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n z_i e_{\text{IV},i} \\ + \frac{1}{\sqrt{n}} \sum_{i=1}^n x_i z_i^\top \Sigma_z^{-1} \mathbb{E}(z_i r_{\text{IV},i}). \end{aligned} \tag{C.7}$$

Thus, in contrast to SV, recentering does not remove the ancillary-variable component but instead produces an additional first-order correction term.

For the first term in (C.7), the consistency argument gives

$$\left(\frac{1}{n} X^\top Z\right) \left(\frac{1}{n} Z^\top Z\right)^{-1} = \Sigma_{xz} \Sigma_z^{-1} + o_p(1).$$

Since $n^{-1/2}\sum_{i=1}^n z_i e_{\mathrm{IV},i} = O_p(1)$, the estimation error in this coefficient contributes only $o_p(1)$.

Hence,

$$\left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n z_i e_{\mathrm{IV},i} = \frac{1}{\sqrt{n}}\sum_{i=1}^n(\Pi_z x_i)e_{\mathrm{IV},i} + o_p(1).$$

Substituting this expression into (C.7) motivates defining

$$\psi_{\mathrm{IV},i} := (\Pi_z x_i)e_{\mathrm{IV},i} + x_i z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i}). \tag{C.8}$$

The normalized numerator then admits the first-order representation

$$\frac{1}{\sqrt{n}}X^\top P_Z r_{\mathrm{IV}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{IV},i} + o_p(1). \tag{C.9}$$

It remains to verify that the summands in (C.9) are centered and have finite second moments. Using $\mathbb{E}(z_i e_{\mathrm{IV},i}) = 0_q$,

$$\mathbb{E}(\psi_{\mathrm{IV},i}) = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i e_{\mathrm{IV},i}) + \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i}) = 0_p,$$

where the last equality follows from (C.2).

For the second moment, $\mathbb{E}\|(\Pi_z x_i)e_{\mathrm{IV},i}\|_2^2 \le \|\Sigma_{xz}\Sigma_z^{-1}\|_{\mathrm{op}}^2\mathbb{E}\|z_i e_{\mathrm{IV},i}\|_2^2 < \infty$. Moreover,

$$\mathbb{E}\left\|x_i z_i^\top \Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i})\right\|_2^2 \le \left\|\Sigma_z^{-1}\mathbb{E}(z_i r_{\mathrm{IV},i})\right\|_2^2\mathbb{E}\|x_i z_i^\top\|_{\mathrm{op}}^2 < \infty.$$

Therefore, $\mathbb{E}\|\psi_{\mathrm{IV},i}\|_2^2 < \infty$. The second bound is precisely where the stronger condition $\mathbb{E}\|x_i z_i^\top\|_{\mathrm{op}}^2 < \infty$ in Assumption 1(ii) is needed. Under valid IV, $\mathbb{E}(z_i r_{\mathrm{IV},i}) = 0_q$, so the correction term in (C.8) vanishes and this additional bound is unnecessary for that term.

Because the observations are i.i.d., the $\psi_{\text{IV},i}$ are i.i.d., centered, and have finite second moments. Therefore, the multivariate central limit theorem gives

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_{\text{IV},i} \xrightarrow{\text{d}} \mathcal{N}(0,\Sigma_{\text{IV}}), \qquad \Sigma_{\text{IV}} := \text{Var}(\psi_{\text{IV},i}). \tag{C.10}$$

Combining (C.9) and (C.10) yields $\frac{1}{\sqrt{n}}X^\top P_Z r_{\text{IV}} \xrightarrow{\text{d}} \mathcal{N}(0,\Sigma_{\text{IV}})$.

Finally, combining this result with (C.5), (C.4), and Slutsky's theorem gives

$$\sqrt{n}(\widehat{\beta}_{\text{IV}} - \beta_{\text{IV}}) \xrightarrow{\text{d}} \mathcal{N}\left(0, Q_{\text{IV}}^{-1}\Sigma_{\text{IV}}Q_{\text{IV}}^{-1}\right) = \mathcal{N}(0, V_{\text{IV}}), \tag{C.11}$$

where $V_{\text{IV}} := Q_{\text{IV}}^{-1}\Sigma_{\text{IV}}Q_{\text{IV}}^{-1}$. This completes the proof.

□

C.2 Proof of Theorem 2

Proof of Theorem 2. Recall that $\Pi_w^\perp x_i = x_i - \Sigma_{xw}\Sigma_w^{-1}w_i$, $Q_{\text{SV}} = \mathbb{E}\{(\Pi_w^\perp x_i)x_i^\top\}$, $\delta_{\text{SV}} = \mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\}$, and $\beta_{\text{SV}} = \beta + Q_{\text{SV}}^{-1}\delta_{\text{SV}}$.

(i) Identification. Recall that the residualized population moment condition is $\mathbb{E}[(\Pi_w^\perp x_i)(y_i - x_i^\top\theta)] = 0_p$, where $\theta \in \mathbb{R}^p$ is a candidate coefficient vector. Using $y_i = x_i^\top\beta + \varsigma_i$, we have

$$\mathbb{E}\Big[(\Pi_w^\perp x_i)\big(y_i - x_i^\top\theta\big)\Big] = \mathbb{E}\Big[(\Pi_w^\perp x_i)x_i^\top\Big](\beta - \theta) + \mathbb{E}\Big[(\Pi_w^\perp x_i)\varsigma_i\Big].$$

By the definitions $Q_{\text{SV}} := \mathbb{E}[(\Pi_w^\perp x_i)x_i^\top]$ and $\delta_{\text{SV}} := \mathbb{E}[(\Pi_w^\perp x_i)\varsigma_i]$, the population moment condition is therefore equivalent to

$$Q_{\text{SV}}(\theta - \beta) = \delta_{\text{SV}}. \tag{C.12}$$

By Assumption 2(i), $Q_{\mathrm{SV}} \in \mathbb{S}^p_{++}$, and hence Q_{SV} is nonsingular. Left-multiplying (C.12) by Q_{SV}^{-1} gives

$$\theta = \beta + Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}} = \beta_{\mathrm{SV}}.$$

Thus a solution exists.

To see that the solution is unique, suppose that both θ_1 and θ_2 satisfy (C.12). Then $Q_{\mathrm{SV}}(\theta_1 - \theta_2) = 0_p$. Since Q_{SV} is nonsingular, its null space contains only 0_p, so $\theta_1 - \theta_2 = 0_p$, and hence $\theta_1 = \theta_2$. Therefore, the residualized population moment uniquely identifies β_{SV}.

Finally, if $\delta_{\mathrm{SV}} = 0_p$, then $Q_{\mathrm{SV}}(\beta_{\mathrm{SV}} - \beta) = 0_p$. Nonsingularity of Q_{SV} implies $\beta_{\mathrm{SV}} = \beta$. Hence, when the residualized moment bias vanishes, the surrogate-control population moment identifies the structural coefficient β.

Within the remainder of the proof, define $\Gamma_w := \Sigma_w^{-1}\Sigma_{wx}$, $\widetilde{x}_i := x_i - \Gamma_w^\top w_i = \Pi_w^\perp x_i$, and let $\widetilde{X}$ denote the matrix whose i-th row is $\widetilde{x}_i^\top$. Then

$$x_i = \Gamma_w^\top w_i + \widetilde{x}_i, \qquad X = W\Gamma_w + \widetilde{X}. \tag{C.13}$$

By the definition of the population linear projection,

$$\mathbb{E}(\widetilde{x}_i w_i^\top) = \Sigma_{xw} - \Gamma_w^\top \Sigma_w = 0_{p\times m}. \tag{C.14}$$

Consequently,

$$Q_{\mathrm{SV}} = \mathbb{E}(\widetilde{x}_i x_i^\top) = \mathbb{E}(\widetilde{x}_i \widetilde{x}_i^\top) + \mathbb{E}(\widetilde{x}_i w_i^\top)\Gamma_w = \mathbb{E}(\widetilde{x}_i \widetilde{x}_i^\top).$$

Hence Q_{SV} is symmetric, and it is positive definite by Assumption 2(i).

Define $r_{\mathrm{SV},i} := y_i - x_i^\top \beta_{\mathrm{SV}} = \varsigma_i - x_i^\top Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}$, and write $r_{\mathrm{SV}} := (r_{\mathrm{SV},1}, \ldots, r_{\mathrm{SV},n})^\top$. Using $\widetilde{x}_i = \Pi_w^\perp x_i$, we obtain

$$\begin{aligned}\mathbb{E}(\widetilde{x}_i r_{\mathrm{SV},i}) &= \mathbb{E}(\widetilde{x}_i \varsigma_i) - \mathbb{E}(\widetilde{x}_i x_i^\top) Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}} \\ &= \delta_{\mathrm{SV}} - Q_{\mathrm{SV}} Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}} \\ &= 0_p.\end{aligned} \tag{C.15}$$

Thus, β_{SV} satisfies the population residualized normal equation.

Since $P_W^\perp W = 0$, decomposition (C.13) gives $P_W^\perp X = P_W^\perp \widetilde{X}$. Because $P_W^\perp$ is symmetric, $X^\top P_W^\perp X = \widetilde{X}^\top P_W^\perp \widetilde{X}$ and $X^\top P_W^\perp r_{\mathrm{SV}} = \widetilde{X}^\top P_W^\perp r_{\mathrm{SV}}$. The estimator therefore satisfies the exact decomposition

$$\widehat{\beta}_{\mathrm{SV}} - \beta_{\mathrm{SV}} = \left(\frac{1}{n}\widetilde{X}^\top P_W^\perp \widetilde{X}\right)^{-1} \left(\frac{1}{n}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}}\right). \tag{C.16}$$

(ii) Consistency. Suppose Assumption 2 (i) holds. The maintained second-moment conditions and the weak law of large numbers give $\frac{1}{n}\widetilde{X}^\top \widetilde{X} \xrightarrow{\mathrm{p}} \mathbb{E}(\widetilde{x}_i \widetilde{x}_i^\top) = Q_{\mathrm{SV}}$, $\frac{1}{n}\widetilde{X}^\top W \xrightarrow{\mathrm{p}} \mathbb{E}(\widetilde{x}_i w_i^\top) = 0_{p\times m}$, and $\frac{1}{n}W^\top W \xrightarrow{\mathrm{p}} \Sigma_w$. Since $\Sigma_w \in \mathbb{S}_{++}^m$,

$$\begin{aligned}\frac{1}{n}\widetilde{X}^\top P_W^\perp \widetilde{X} &= \frac{1}{n}\widetilde{X}^\top \widetilde{X} - \left(\frac{1}{n}\widetilde{X}^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\left(\frac{1}{n}W^\top \widetilde{X}\right) \\ &\xrightarrow{\mathrm{p}} Q_{\mathrm{SV}}.\end{aligned}$$

It follows that $\{\frac{1}{n}\widetilde{X}^\top P_W^\perp \widetilde{X}\}^{-1} \xrightarrow{\mathrm{p}} Q_{\mathrm{SV}}^{-1}$.

Similarly, the weak law of large numbers gives $\frac{1}{n}\sum_{i=1}^n \widetilde{x}_i r_{\mathrm{SV},i} \xrightarrow{\mathrm{p}} \mathbb{E}(\widetilde{x}_i r_{\mathrm{SV},i}) = 0_p$ by (C.15), and $\frac{1}{n}\sum_{i=1}^n w_i r_{\mathrm{SV},i} \xrightarrow{\mathrm{p}} \mathbb{E}(w_i r_{\mathrm{SV},i})$. Therefore,

$$\begin{aligned}\frac{1}{n}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}} &= \frac{1}{n}\sum_{i=1}^n \widetilde{x}_i r_{\mathrm{SV},i} - \left(\frac{1}{n}\widetilde{X}^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^n w_i r_{\mathrm{SV},i}\right) \\ &\xrightarrow{\mathrm{p}} 0_p.\end{aligned}$$

Combining these limits with (C.16) yields

$$\widehat{\beta}_{\mathrm{SV}} \xrightarrow{\mathrm{p}} \beta_{\mathrm{SV}}.$$

(iii) Asymptotic normality. Suppose further that Assumption 2 (ii) holds. Multiplying (C.16) by $\sqrt{n}$ gives

$$\sqrt{n}(\widehat{\beta}_{\mathrm{SV}} - \beta_{\mathrm{SV}}) = \left(\frac{1}{n}\widetilde{X}^\top P_W^\perp \widetilde{X}\right)^{-1} \frac{1}{\sqrt{n}}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}}.$$

The first factor converges in probability to Q_{SV}^{-1}, so it remains to derive the limiting distribution of the normalized numerator.

A direct expansion gives

$$\frac{1}{\sqrt{n}}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n \widetilde{x}_i r_{\mathrm{SV},i} - \left(\frac{1}{n}\widetilde{X}^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n w_i r_{\mathrm{SV},i}. \tag{C.17}$$

Although $\mathbb{E}(\widetilde{x}_i r_{\mathrm{SV},i}) = 0_p$, this does not imply $\mathbb{E}(w_i r_{\mathrm{SV},i}) = 0_m$. Hence the last sum in (C.17) may contain a nonzero mean component of order $\sqrt{n}$, so the second term cannot yet be treated as negligible.

To center this sum, define

$$c_{\mathrm{SV}} := \Sigma_w^{-1}\mathbb{E}(w_i r_{\mathrm{SV},i}), \qquad e_{\mathrm{SV},i} := r_{\mathrm{SV},i} - w_i^\top c_{\mathrm{SV}}. \tag{C.18}$$

Then $\mathbb{E}(w_i e_{\mathrm{SV},i}) = \mathbb{E}(w_i r_{\mathrm{SV},i}) - \Sigma_w c_{\mathrm{SV}} = 0_m$. Writing $e_{\mathrm{SV}} := (e_{\mathrm{SV},1}, \ldots, e_{\mathrm{SV},n})^\top$, equation (C.18) gives $r_{\mathrm{SV}} = e_{\mathrm{SV}} + W c_{\mathrm{SV}}$. Since $P_W^\perp W = 0$, we have the exact identity

$$P_W^\perp r_{\mathrm{SV}} = P_W^\perp e_{\mathrm{SV}}.$$

The corrected numerator therefore satisfies

$$\frac{1}{\sqrt{n}}\widetilde{X}^\top P_W^\perp r_{\mathrm{SV}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n \widetilde{x}_i e_{\mathrm{SV},i} - \left(\frac{1}{n}\widetilde{X}^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^n w_i e_{\mathrm{SV},i}.$$

Assumption 2(ii) and $\mathbb{E}(w_i e_{\mathrm{SV},i}) = 0_m$ imply $n^{-1/2}\sum_{i=1}^n w_i e_{\mathrm{SV},i} = O_p(1)$ by the multivariate central limit theorem. Moreover, $\frac{1}{n}\widetilde{X}^\top W = o_p(1)$ and $\{\frac{1}{n}W^\top W\}^{-1} = O_p(1)$. Thus, the second term is $o_p(1)$.

Define $\psi_{\mathrm{SV},i} := \widetilde{x}_i e_{\mathrm{SV},i} = (\Pi_w^\perp x_i)e_{\mathrm{SV},i}$. The preceding argument gives the first-order representation

$$\frac{1}{\sqrt{n}}X^\top P_W^\perp r_{\mathrm{SV}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n \psi_{\mathrm{SV},i} + o_p(1). \tag{C.19}$$

The score is centered because, by (C.14) and (C.15),

$$\mathbb{E}(\psi_{\mathrm{SV},i}) = \mathbb{E}(\widetilde{x}_i r_{\mathrm{SV},i}) - \mathbb{E}(\widetilde{x}_i w_i^\top)c_{\mathrm{SV}} = 0_p.$$

It remains to verify that $\mathbb{E}\|\psi_{\mathrm{SV},i}\|_2^2 < \infty$. Since $r_{\mathrm{SV},i} = \varsigma_i - x_i^\top Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}$,

$$\mathbb{E}\|x_i r_{\mathrm{SV},i}\|_2^2 \le 2\mathbb{E}\|x_i\varsigma_i\|_2^2 + 2\|Q_{\mathrm{SV}}^{-1}\delta_{\mathrm{SV}}\|_2^2\mathbb{E}\|x_i x_i^\top\|_{\mathrm{op}}^2 < \infty.$$

The consistency argument and the nonsingularity of Σ_w imply that c_{SV} in (C.18) is finite. Hence,

$$\mathbb{E}\|x_i e_{\mathrm{SV},i}\|_2^2 \le 2\mathbb{E}\|x_i r_{\mathrm{SV},i}\|_2^2 + 2\|c_{\mathrm{SV}}\|_2^2\mathbb{E}\|x_i w_i^\top\|_{\mathrm{op}}^2 < \infty.$$

Finally, $\psi_{\mathrm{SV},i} = x_i e_{\mathrm{SV},i} - \Gamma_w^\top w_i e_{\mathrm{SV},i}$, so

$$\mathbb{E}\|\psi_{\mathrm{SV},i}\|_2^2 \le 2\mathbb{E}\|x_i e_{\mathrm{SV},i}\|_2^2 + 2\|\Gamma_w^\top\|_{\mathrm{op}}^2\mathbb{E}\|w_i e_{\mathrm{SV},i}\|_2^2 < \infty.$$

Because the observations are i.i.d., the random vectors $\psi_{\text{SV},i}$ are i.i.d., centered, and have finite second moments. The multivariate central limit theorem therefore gives

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_{\text{SV},i} \xrightarrow{\text{d}} \mathcal{N}(0,\Sigma_{\text{SV}}), \qquad \Sigma_{\text{SV}} := \text{Var}(\psi_{\text{SV},i}).$$

Combining this result with (C.19) yields

$$\frac{1}{\sqrt{n}}X^{\top}P_W^{\perp}r_{\text{SV}} \xrightarrow{\text{d}} \mathcal{N}(0,\Sigma_{\text{SV}}).$$

Finally, Slutsky's theorem gives

$$\sqrt{n}(\widehat{\beta}_{\text{SV}}-\beta_{\text{SV}}) \xrightarrow{\text{d}} \mathcal{N}\left(0, Q_{\text{SV}}^{-1}\Sigma_{\text{SV}}Q_{\text{SV}}^{-1}\right).$$

Thus,

$$V_{\text{SV}} = Q_{\text{SV}}^{-1}\Sigma_{\text{SV}}Q_{\text{SV}}^{-1}.$$

This completes the proof.

□

C.3 Proof of Proposition 3

Proof of Proposition 3. Recall that $h_i = \{(\Pi_z x_i)^{\top}, (\Pi_w^{\perp}x_i)^{\top}\}^{\top}$ and $J = \mathbb{E}(h_i x_i^{\top}) = \{Q_{\text{IV}}^{\top}, Q_{\text{SV}}^{\top}\}^{\top}$. Under exact projected validity, $\mathbb{E}(h_i\varsigma_i) = 0_{2p}$.

Population moment equation. Let $\theta \in \mathbb{R}^p$ be a candidate coefficient vector. Using $y_i = x_i^{\top}\beta + \varsigma_i$, the stacked population moment satisfies

$$\mathbb{E}\{h_i(y_i - x_i^{\top}\theta)\} = J(\beta-\theta) + \mathbb{E}(h_i\varsigma_i) = J(\beta-\theta).$$

Hence, θ satisfies the stacked moment restriction if and only if $J(\theta - \beta) = 0_{2p}$. Therefore, β is the unique solution if and only if $\ker(J) = \{0_p\}$, which is equivalent to $\operatorname{rank}(J) = p$.

Rank characterization. Because J stacks the two projected Jacobian blocks, $Jv = 0_{2p}$ for $v \in \mathbb{R}^p$ if and only if $Q_{\text{IV}}v = 0_p$ and $Q_{\text{SV}}v = 0_p$. It follows that

$$\ker(J) = \ker(Q_{\text{IV}}) \cap \ker(Q_{\text{SV}}).$$

Consequently, $\operatorname{rank}(J) = p$ if and only if $\ker(Q_{\text{IV}}) \cap \ker(Q_{\text{SV}}) = \{0_p\}$, proving (3.2).

Complementarity. The preceding characterization does not require either $\ker(Q_{\text{IV}})$ or $\ker(Q_{\text{SV}})$ to be trivial individually. Thus, either projected moment block may be rank deficient as long as no nonzero parameter direction belongs to both null spaces. Equivalently, directions that are not identified by one block may still be identified by the other, and the stacked system uniquely identifies β whenever their unidentified directions have trivial intersection.

□

C.4 Proof of Theorem 4

Proof of Theorem 4. Recall that $\varsigma_i = y_i - x_i^\top \beta$ and

$$\widehat{g}(\theta) = \frac{1}{n} \begin{pmatrix} X^\top P_Z (Y - X\theta) \\ \\ X^\top P_W^\perp (Y - X\theta) \end{pmatrix}.$$

Recall that

$$\widehat{J} = \frac{1}{n} \begin{pmatrix} X^\top P_Z X \\ \\ X^\top P_W^\perp X \end{pmatrix}.$$

Since the sample moment is linear in θ,

$$\widehat{g}(\theta) = \widehat{g}(\beta) - \widehat{J}(\theta - \beta). \tag{C.20}$$

Assumption 3 (i) gives $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$ and $\mathbb{E}\{(\Pi_w^{\perp} x_i)\varsigma_i\} = 0_p$. Thus, both projected moment blocks are correctly specified at β.

Identification. Let $g(\theta)$ denote the population counterpart of $\widehat{g}(\theta)$. By linearity and the projected moment restrictions,

$$g(\theta) = -J(\theta - \beta).$$

Hence $g(\theta) = 0_{2p}$ if and only if $J(\theta - \beta) = 0_{2p}$. Since Assumption 3(i) gives $\operatorname{rank}(J) = p$, the null space of J is trivial. Therefore, $\theta = \beta$, so β is the unique solution to the population combined projected moment condition.

The first-order condition for $\widehat{\beta}(M)$ is $\widehat{J}^{\top} M \widehat{g}\{\widehat{\beta}(M)\} = 0_p$. Using (C.20), we obtain the exact estimation-error decomposition

$$\widehat{\beta}(M) - \beta = \left(\widehat{J}^{\top} M \widehat{J}\right)^{-1} \widehat{J}^{\top} M \widehat{g}(\beta). \tag{C.21}$$

Consistency. Suppose Assumption 3 (i) holds. We first establish the probability limit of $\widehat{J}$.

For the IV block, the maintained second-moment conditions and the weak law of large numbers give $\frac{1}{n} X^{\top} Z \xrightarrow{\mathrm{p}} \Sigma_{xz}$, $\frac{1}{n} Z^{\top} Z \xrightarrow{\mathrm{p}} \Sigma_z$, and $\frac{1}{n} Z^{\top} X \xrightarrow{\mathrm{p}} \Sigma_{zx}$. Since $\Sigma_z \in \mathbb{S}_{++}^q$, $(\frac{1}{n} Z^{\top} Z)^{-1} \xrightarrow{\mathrm{p}}$

Σ_z^{-1}. Hence,

$$\frac{1}{n}X^\top P_Z X = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\left(\frac{1}{n}Z^\top X\right) \xrightarrow{\text{p}} Q_{\text{IV}}.$$

For the SV block, the weak law of large numbers gives $\frac{1}{n}X^\top X \xrightarrow{\text{p}} \Sigma_x$, $\frac{1}{n}X^\top W \xrightarrow{\text{p}} \Sigma_{xw}$, $\frac{1}{n}W^\top W \xrightarrow{\text{p}} \Sigma_w$, and $\frac{1}{n}W^\top X \xrightarrow{\text{p}} \Sigma_{wx}$. Since $\Sigma_w \in \mathbb{S}^m_{++}$, $(\frac{1}{n}W^\top W)^{-1} \xrightarrow{\text{p}} \Sigma_w^{-1}$. Therefore,

$$\frac{1}{n}X^\top P_W^\perp X \xrightarrow{\text{p}} \Sigma_x - \Sigma_{xw}\Sigma_w^{-1}\Sigma_{wx} = Q_{\text{SV}}.$$

Combining the two blocks gives

$$\widehat{J} \xrightarrow{\text{p}} \begin{pmatrix} Q_{\text{IV}} \\ \\ Q_{\text{SV}} \end{pmatrix} = J. \tag{C.22}$$

We next establish the probability limit of $\widehat{g}(\beta)$. For the IV block,

$$\frac{1}{n}X^\top P_Z \varsigma = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^n z_i\varsigma_i\right) \xrightarrow{\text{p}} \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i) = 0_p,$$

where the final equality follows from $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$. In particular, this does not require the stronger raw IV restriction $\mathbb{E}(z_i\varsigma_i) = 0_q$.

For the SV block,

$$\frac{1}{n}X^\top P_W^\perp \varsigma \xrightarrow{\text{p}} \mathbb{E}(x_i\varsigma_i) - \Sigma_{xw}\Sigma_w^{-1}\mathbb{E}(w_i\varsigma_i) = \mathbb{E}\{(\Pi_w^\perp x_i)\varsigma_i\} = 0_p.$$

Hence,

$$\widehat{g}(\beta) \xrightarrow{\text{p}} 0_{2p}. \tag{C.23}$$

By Assumption 3 (i), $J^\top M J \in \mathbb{S}^p_{++}$. Together with (C.22), this implies that $\widehat{J}^\top M \widehat{J}$ is positive definite with probability approaching one. The continuous mapping theorem therefore

gives

$$\left(\widehat{J}^\top M \widehat{J}\right)^{-1} \widehat{J}^\top M \xrightarrow{\mathrm{p}} (J^\top M J)^{-1} J^\top M = K_M.$$

Combining this result with (C.21) and (C.23) gives $\widehat{\beta}(M) - \beta \xrightarrow{\mathrm{p}} 0_p$. Therefore, $\widehat{\beta}(M) \xrightarrow{\mathrm{p}} \beta$.

Asymptotic normality. Suppose further that Assumption 3 (ii) holds. Multiplying (C.21) by $\sqrt{n}$ gives

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} = \left(\widehat{J}^\top M \widehat{J}\right)^{-1} \widehat{J}^\top M \sqrt{n}\, \widehat{g}(\beta). \tag{C.24}$$

By the consistency argument, the matrix multiplying $\sqrt{n}\,\widehat{g}(\beta)$ converges to K_M. It therefore remains to derive a central limit theorem for the two projected moment blocks in $\sqrt{n}\,\widehat{g}(\beta)$.

Define the population projection residuals

$$e_{z,i} := \varsigma_i - z_i^\top \Sigma_z^{-1} \mathbb{E}(z_i \varsigma_i), \qquad e_{w,i} := \varsigma_i - w_i^\top \Sigma_w^{-1} \mathbb{E}(w_i \varsigma_i).$$

The maintained second-moment conditions ensure that the disturbance cross moments are finite. By construction,

$$\mathbb{E}(z_i e_{z,i}) = 0_q, \qquad \mathbb{E}(w_i e_{w,i}) = 0_m.$$

The standing second-moment conditions imply that $\mathbb{E}(z_i\varsigma_i)$ and $\mathbb{E}(w_i\varsigma_i)$ are finite, so c_z and c_w are finite nonrandom vectors. By construction, $\mathbb{E}(z_i e_{z,i}) = 0_q$ and $\mathbb{E}(w_i e_{w,i}) = 0_m$.

We first consider the projected IV block. A direct expansion in terms of ς_i cannot generally be treated by replacing the sample projection coefficient by its probability limit, because the projected restriction $\Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i) = 0_p$ does not imply $\mathbb{E}(z_i\varsigma_i) = 0_q$. Thus, $n^{-1/2}\sum_{i=1}^n z_i\varsigma_i$ may contain a mean component of order $\sqrt{n}$.

Using $\varsigma_i = e_{z,i} + z_i^\top c_z$, the IV block admits the exact decomposition

$$\frac{1}{\sqrt{n}} X^\top P_Z \varsigma = \left(\frac{1}{n} X^\top Z\right)\left(\frac{1}{n} Z^\top Z\right)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n z_i e_{z,i} + \frac{1}{\sqrt{n}} \sum_{i=1}^n x_i z_i^\top c_z. \tag{C.25}$$

The second term remains because $P_Z Z = Z$; hence, the component of the structural error lying in the span of Z is retained by the IV projection.

Assumption 3 (ii) gives $\mathbb{E}\|z_i e_{z,i}\|_2^2 < \infty$, so $n^{-1/2} \sum_{i=1}^n z_i e_{z,i} = O_p(1)$. Moreover,

$$\left(\frac{1}{n} X^\top Z\right)\left(\frac{1}{n} Z^\top Z\right)^{-1} = \Sigma_{xz}\Sigma_z^{-1} + o_p(1).$$

Therefore, the first term in (C.25) equals $n^{-1/2} \sum_{i=1}^n (\Pi_z x_i) e_{z,i} + o_p(1)$, and hence

$$\frac{1}{\sqrt{n}} X^\top P_Z \varsigma = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left\{(\Pi_z x_i) e_{z,i} + x_i z_i^\top c_z\right\} + o_p(1). \tag{C.26}$$

We next consider the projected SV block. Here the same recentering has a different consequence. Since $\varsigma_i = e_{w,i} + w_i^\top c_w$ and $P_W^\perp W = 0$, the component lying in the span of W is removed exactly:

$$\frac{1}{\sqrt{n}} X^\top P_W^\perp \varsigma = \frac{1}{\sqrt{n}} X^\top P_W^\perp e_w,$$

where $e_w = (e_{w,1}, \ldots, e_{w,n})^\top$.

Let $\Pi_w^\perp X$ denote the matrix whose i-th row is $(\Pi_w^\perp x_i)^\top$. Since $X = \Pi_w^\perp X + W\Sigma_w^{-1}\Sigma_{wx}$, the identity $P_W^\perp W = 0$ implies $P_W^\perp X = P_W^\perp(\Pi_w^\perp X)$. Expanding the sample residual-maker therefore gives

$$\frac{1}{\sqrt{n}} X^\top P_W^\perp \varsigma = \frac{1}{\sqrt{n}} \sum_{i=1}^n (\Pi_w^\perp x_i) e_{w,i}$$

$$-\left\{\frac{1}{n}\sum_{i=1}^{n}(\Pi_w^{\perp}x_i)w_i^{\top}\right\}\left(\frac{1}{n}\sum_{i=1}^{n}w_iw_i^{\top}\right)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n}w_ie_{w,i}. \tag{C.27}$$

By the definition of the population residualization, $\mathbb{E}\{(\Pi_w^{\perp}x_i)w_i^{\top}\} = 0_{p\times m}$. The weak law of large numbers therefore gives $n^{-1}\sum_{i=1}^{n}(\Pi_w^{\perp}x_i)w_i^{\top} = o_p(1)$. Also, $(n^{-1}\sum_{i=1}^{n}w_iw_i^{\top})^{-1} \xrightarrow{\text{P}} \Sigma_w^{-1}$, while $\mathbb{E}(w_ie_{w,i}) = 0_m$ and $\mathbb{E}\|w_ie_{w,i}\|_2^2 < \infty$ imply $n^{-1/2}\sum_{i=1}^{n}w_ie_{w,i} = O_p(1)$. Hence, the second term in (C.27) is $o_p(1)$, so

$$\frac{1}{\sqrt{n}}X^{\top}P_W^{\perp}\varsigma = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}(\Pi_w^{\perp}x_i)e_{w,i} + o_p(1). \tag{C.28}$$

Equations (C.26) and (C.28) collect the first-order variation of the two feasible projected moment blocks. Recall from Section 3.1 that $\psi_{\text{IV},i}$ and $\psi_{\text{SV},i}$ denote the corresponding first-order scores. Under the notation introduced above, these previously defined scores admit the equivalent representations

$$\psi_{\text{IV},i} = (\Pi_z x_i)e_{z,i} + x_iz_i^{\top}c_z, \qquad \psi_{\text{SV},i} = (\Pi_w^{\perp}x_i)e_{w,i}.$$

Recall also that $\psi_i = (\psi_{\text{IV},i}^{\top}, \psi_{\text{SV},i}^{\top})^{\top}$ and $\Sigma = \text{Var}(\psi_i)$. Stacking (C.26) and (C.28) gives the first-order representation

$$\sqrt{n}\,\widehat{g}(\beta) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_i + o_p(1). \tag{C.29}$$

It remains to verify that the summands in (C.29) are centered and have finite second moments.

For the IV component, using $\mathbb{E}(z_ie_{z,i}) = 0_q$,

$$\mathbb{E}(\psi_{\text{IV},i}) = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_ie_{z,i}) + \Sigma_{xz}c_z = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i) = 0_p,$$

where the last equality follows from the projected IV restriction. For the SV component,

$$\mathbb{E}(\psi_{\mathrm{SV},i}) = \mathbb{E}\{(\Pi_w^{\perp} x_i)\varsigma_i\} - \mathbb{E}\{(\Pi_w^{\perp} x_i) w_i^{\top}\} c_w = 0_p.$$

Hence,

$$\mathbb{E}(\psi_i) = \begin{pmatrix} \mathbb{E}(\psi_{\mathrm{IV},i}) \\ \\ \mathbb{E}(\psi_{\mathrm{SV},i}) \end{pmatrix} = \begin{pmatrix} 0_p \\ \\ 0_p \end{pmatrix} = 0_{2p}.$$

For the IV second moment, $\mathbb{E}\|(\Pi_z x_i) e_{z,i}\|_2^2 \le \|\Sigma_{xz}\Sigma_z^{-1}\|_{\mathrm{op}}^2 \mathbb{E}\|z_i e_{z,i}\|_2^2 < \infty$. Moreover, $\mathbb{E}\|x_i z_i^{\top} c_z\|_2^2 \le \|c_z\|_2^2 \mathbb{E}\|x_i z_i^{\top}\|_{\mathrm{op}}^2 < \infty$. Thus, $\mathbb{E}\|\psi_{\mathrm{IV},i}\|_2^2 < \infty$.

For the SV component, first note that $x_i e_{w,i} = x_i \varsigma_i - x_i w_i^{\top} c_w$. Therefore,

$$\mathbb{E}\|x_i e_{w,i}\|_2^2 \le 2\mathbb{E}\|x_i \varsigma_i\|_2^2 + 2\|c_w\|_2^2 \mathbb{E}\|x_i w_i^{\top}\|_{\mathrm{op}}^2 < \infty.$$

Since $(\Pi_w^{\perp} x_i) e_{w,i} = x_i e_{w,i} - \Sigma_{xw}\Sigma_w^{-1} w_i e_{w,i}$, we further have

$$\mathbb{E}\|\psi_{\mathrm{SV},i}\|_2^2 \le 2\mathbb{E}\|x_i e_{w,i}\|_2^2 + 2\|\Sigma_{xw}\Sigma_w^{-1}\|_{\mathrm{op}}^2 \mathbb{E}\|w_i e_{w,i}\|_2^2 < \infty.$$

Consequently, $\mathbb{E}\|\psi_i\|_2^2 < \infty$, so $\Sigma = \mathrm{Var}(\psi_i)$ is well defined.

Because the observations are i.i.d., the ψ_i are i.i.d., centered, and have finite second moments. The multivariate central limit theorem therefore gives

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_i \xrightarrow{\mathrm{d}} \mathcal{N}(0, \Sigma). \tag{C.30}$$

Combining (C.29) and (C.30) yields

$$\sqrt{n}\, \widehat{g}(\beta) \xrightarrow{\mathrm{d}} \mathcal{N}(0, \Sigma).$$

Finally, by (C.22),

$$\left(\widehat{J}^\top M \widehat{J}\right)^{-1} \widehat{J}^\top M \xrightarrow{\mathrm{p}} (J^\top M J)^{-1} J^\top M = K_M.$$

Combining this result with (C.24) and Slutsky's theorem gives

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\mathrm{d}} \mathcal{N}\left(0,\, K_M \Sigma K_M^\top\right) = \mathcal{N}\{0, V(M)\}, \tag{C.31}$$

where $V(M) = K_M \Sigma K_M^\top$. By Assumption 3 (ii), $V(M) \in \mathbb{S}^p_{++}$, so the limiting distribution is nondegenerate. This completes the proof.

□

C.5 Proof of Theorem 5

Proof of Theorem 5. Recall that

$$\widehat{g}(\theta) = \frac{1}{n} \begin{pmatrix} X^\top P_Z (Y - X\theta) \\ \\ X^\top P_W^\perp (Y - X\theta) \end{pmatrix}, \qquad \widehat{J} = \frac{1}{n} \begin{pmatrix} X^\top P_Z X \\ \\ X^\top P_W^\perp X \end{pmatrix}.$$

Since the sample moment is linear in θ,

$$\widehat{g}(\theta) = \widehat{g}(\beta) - \widehat{J}(\theta - \beta).$$

Throughout the local sequence, the joint distribution of (x_i, z_i, w_i) is fixed. Hence the population projections, design second moments, J, and K_M do not depend on n. Expectations involving ς_i are taken under the row-n distribution and denoted by $\mathbb{E}_n$.

Population target. Let $g_n(\theta) := \mathbb{E}_n\{h_i(y_i - x_i^\top\theta)\}$. Using $y_i = x_i^\top\beta + \varsigma_i$ and (3.6),

$$g_n(\theta) = \frac{\tau}{\sqrt{n}} - J(\theta - \beta).$$

The population GMM criterion is

$$g_n(\theta)^\top M g_n(\theta).$$

By Assumption 4 (i), $J^\top M J \in \mathbb{S}^p_{++}$, so the criterion is strictly convex. Its unique minimizer therefore satisfies

$$J^\top M\left\{\frac{\tau}{\sqrt{n}} - J(\beta_{M,n} - \beta)\right\} = 0_p.$$

Hence

$$\beta_{M,n} - \beta = \frac{1}{\sqrt{n}}(J^\top M J)^{-1} J^\top M\tau = \frac{1}{\sqrt{n}} K_M \tau,$$

and therefore

$$\beta_{M,n} = \beta + n^{-1/2} K_M \tau \longrightarrow \beta.$$

Consistency. Suppose further that Assumption 4 (ii) holds. The observations are i.i.d. within each row, and the assumed second moments imply, by Chebyshev's inequality applied entrywise,

$$\frac{1}{n}X^\top Z = \Sigma_{xz} + O_p(n^{-1/2}), \qquad \frac{1}{n}Z^\top Z = \Sigma_z + O_p(n^{-1/2}),$$

$$\frac{1}{n}X^\top X = \Sigma_x + O_p(n^{-1/2}), \qquad \frac{1}{n}X^\top W = \Sigma_{xw} + O_p(n^{-1/2}),$$

$$\frac{1}{n}W^\top W = \Sigma_w + O_p(n^{-1/2}).$$

Since Σ_z and Σ_w are positive definite,

$$\left(\frac{1}{n}Z^\top Z\right)^{-1} \xrightarrow{\text{p}} \Sigma_z^{-1}, \qquad \left(\frac{1}{n}W^\top W\right)^{-1} \xrightarrow{\text{p}} \Sigma_w^{-1}.$$

It follows that

$$\frac{1}{n}X^\top P_Z X = \left(\frac{1}{n}X^\top Z\right)\left(\frac{1}{n}Z^\top Z\right)^{-1}\left(\frac{1}{n}Z^\top X\right) \xrightarrow{\text{p}} Q_{\text{IV}},$$

and

$$\frac{1}{n}X^\top P_W^\perp X = \frac{1}{n}X^\top X - \left(\frac{1}{n}X^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1}\left(\frac{1}{n}W^\top X\right)$$

$$\xrightarrow{\text{p}} \Sigma_x - \Sigma_{xw}\Sigma_w^{-1}\Sigma_{wx} = Q_{\text{SV}}.$$

Therefore,

$$\widehat{J} \xrightarrow{\text{p}} \begin{pmatrix} Q_{\text{IV}} \\ \\ Q_{\text{SV}} \end{pmatrix} = J.$$

Since $J^\top M J \in \mathbb{S}^p_{++}$, the matrix $\widehat{J}^\top M \widehat{J}$ is nonsingular with probability approaching one, and

$$(\widehat{J}^\top M \widehat{J})^{-1}\widehat{J}^\top M \xrightarrow{\text{p}} (J^\top M J)^{-1} J^\top M = K_M. \tag{C.32}$$

It remains to control the sample moments at β. Define

$$e_{z,i} := \varsigma_i - z_i^\top \Sigma_z^{-1}\mathbb{E}_n(z_i\varsigma_i), \qquad e_{w,i} := \varsigma_i - w_i^\top \Sigma_w^{-1}\mathbb{E}_n(w_i\varsigma_i).$$

By construction,

$$\mathbb{E}_n(z_i e_{z,i}) = 0_q, \qquad \mathbb{E}_n(w_i e_{w,i}) = 0_m.$$

Cauchy–Schwarz and the fixed design second moments give

$$\sup_n \|\mathbb{E}_n(z_i\varsigma_i)\|_2 < \infty, \qquad \sup_n \|\mathbb{E}_n(w_i\varsigma_i)\|_2 < \infty.$$

Moreover,

$$z_i\varsigma_i = z_i e_{z,i} + z_i z_i^\top \Sigma_z^{-1}\mathbb{E}_n(z_i\varsigma_i),$$

so Assumption 4 (ii) implies

$$\sup_n \mathbb{E}_n \|z_i \varsigma_i\|_2^2 < \infty.$$

The same argument gives

$$\sup_n \mathbb{E}_n \|w_i \varsigma_i\|_2^2 < \infty.$$

Together with $\sup_n \mathbb{E}_n \|x_i \varsigma_i\|_2^2 < \infty$, Chebyshev's inequality yields

$$\frac{1}{n} X^\top \varsigma = \mathbb{E}_n(x_i \varsigma_i) + O_p(n^{-1/2}),$$

$$\frac{1}{n} Z^\top \varsigma = \mathbb{E}_n(z_i \varsigma_i) + O_p(n^{-1/2}),$$

$$\frac{1}{n} W^\top \varsigma = \mathbb{E}_n(w_i \varsigma_i) + O_p(n^{-1/2}).$$

Using

$$\frac{1}{n} X^\top P_Z \varsigma = \left(\frac{1}{n} X^\top Z\right) \left(\frac{1}{n} Z^\top Z\right)^{-1} \left(\frac{1}{n} Z^\top \varsigma\right)$$

and

$$\frac{1}{n} X^\top P_W^\perp \varsigma = \frac{1}{n} X^\top \varsigma - \left(\frac{1}{n} X^\top W\right) \left(\frac{1}{n} W^\top W\right)^{-1} \left(\frac{1}{n} W^\top \varsigma\right),$$

we obtain

$$\widehat{g}(\beta) - g_n(\beta) = O_p(n^{-1/2}).$$

Since

$$g_n(\beta) = \frac{\tau}{\sqrt{n}},$$

it follows that

$$\widehat{g}(\beta) = O_p(n^{-1/2}) \xrightarrow{\mathrm{p}} 0_{2p}.$$

The first-order condition for $\widehat{\beta}(M)$, together with the linearity of $\widehat{g}(\theta)$, gives the exact decomposition

$$\widehat{\beta}(M) - \beta = (\widehat{J}^\top M \widehat{J})^{-1} \widehat{J}^\top M \widehat{g}(\beta). \tag{C.33}$$

Combining (C.32) with $\widehat{g}(\beta) \xrightarrow{\mathrm{p}} 0_{2p}$ gives

$$\widehat{\beta}(M) \xrightarrow{\mathrm{p}} \beta.$$

Asymptotic normality. Suppose further that Assumption 4 (iii) holds. Multiplying (C.33) by $\sqrt{n}$,

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} = (\widehat{J}^\top M \widehat{J})^{-1} \widehat{J}^\top M \sqrt{n}\, \widehat{g}(\beta).$$

By (C.32), the matrix multiplying $\sqrt{n}\, \widehat{g}(\beta)$ converges in probability to K_M. It therefore remains to derive the limiting distribution of $\sqrt{n}\, \widehat{g}(\beta)$.

For the IV block, the score in (3.5) can be written as

$$\psi_{\mathrm{IV},i} = \Sigma_{xz} \Sigma_z^{-1} z_i e_{z,i} + x_i z_i^\top \Sigma_z^{-1} \mathbb{E}_n(z_i \varsigma_i).$$

A direct calculation gives

$$\begin{aligned} &\frac{1}{\sqrt{n}} X^\top P_Z \varsigma - \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_{\mathrm{IV},i} \\ &\quad = \left\{ \left(\frac{1}{n} X^\top Z \right) \left(\frac{1}{n} Z^\top Z \right)^{-1} - \Sigma_{xz} \Sigma_z^{-1} \right\} \frac{1}{\sqrt{n}} \sum_{i=1}^n z_i e_{z,i}. \end{aligned}$$

The matrix in braces is $o_p(1)$. Moreover,

$$\mathbb{E}_n(z_i e_{z,i}) = 0_q, \qquad \sup_n \mathbb{E}_n \| z_i e_{z,i} \|_2^2 < \infty,$$

so

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n} z_i e_{z,i} = O_p(1).$$

Hence

$$\frac{1}{\sqrt{n}}X^\top P_Z \varsigma = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_{\text{IV},i} + o_p(1).$$

For the SV block, $P_W^\perp W = 0$ gives

$$X^\top P_W^\perp \varsigma = X^\top P_W^\perp e_w,$$

where $e_w = (e_{w,1}, \ldots, e_{w,n})^\top$. Since

$$\psi_{\text{SV},i} = (x_i - \Sigma_{xw}\Sigma_w^{-1} w_i) e_{w,i},$$

we similarly obtain

$$\begin{aligned}
&\frac{1}{\sqrt{n}}X^\top P_W^\perp \varsigma - \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_{\text{SV},i} \\
&\quad = -\left\{\left(\frac{1}{n}X^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1} - \Sigma_{xw}\Sigma_w^{-1}\right\}\frac{1}{\sqrt{n}}\sum_{i=1}^{n} w_i e_{w,i}.
\end{aligned}$$

Again,

$$\left(\frac{1}{n}X^\top W\right)\left(\frac{1}{n}W^\top W\right)^{-1} - \Sigma_{xw}\Sigma_w^{-1} = o_p(1),$$

while

$$\mathbb{E}_n(w_i e_{w,i}) = 0_m, \qquad \frac{1}{\sqrt{n}}\sum_{i=1}^{n} w_i e_{w,i} = O_p(1).$$

Therefore,

$$\frac{1}{\sqrt{n}}X^\top P_W^\perp \varsigma = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_{\text{SV},i} + o_p(1).$$

Stacking the two blocks gives

$$\sqrt{n}\,\widehat{g}(\beta) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_i + o_p(1). \tag{C.34}$$

We next determine the mean of the score. Using

$$\mathbb{E}\{(x_i - \Pi_z x_i)z_i^\top\} = 0_{p\times q}, \qquad \mathbb{E}\{(\Pi_w^\perp x_i)w_i^\top\} = 0_{p\times m},$$

we have

$$\mathbb{E}_n(\psi_{\text{IV},i}) = \mathbb{E}_n\{(\Pi_z x_i)\varsigma_i\}, \qquad \mathbb{E}_n(\psi_{\text{SV},i}) = \mathbb{E}_n\{(\Pi_w^\perp x_i)\varsigma_i\}.$$

Hence, by (3.6),

$$\mathbb{E}_n(\psi_i) = \mathbb{E}_n(h_i\varsigma_i) = \frac{\tau}{\sqrt{n}}.$$

Consequently,

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_i = \tau + \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\{\psi_i - \mathbb{E}_n(\psi_i)\}.$$

It remains to establish a central limit theorem for the centered triangular array. Assumption 4 (iii) gives, for some $\eta > 0$,

$$\sup_n \mathbb{E}_n\|\psi_i\|_2^{2+\eta} < \infty.$$

Jensen's inequality therefore implies

$$\sup_n \mathbb{E}_n\|\psi_i - \mathbb{E}_n(\psi_i)\|_2^{2+\eta} < \infty.$$

For every $\epsilon > 0$,

$$\mathbb{E}_n\left[\|\psi_i - \mathbb{E}_n(\psi_i)\|_2^2\mathbf{1}\left\{\|\psi_i - \mathbb{E}_n(\psi_i)\|_2 > \epsilon\sqrt{n}\right\}\right]$$

$$\leq (\epsilon\sqrt{n})^{-\eta}\mathbb{E}_n\|\psi_i - \mathbb{E}_n(\psi_i)\|_2^{2+\eta} \longrightarrow 0.$$

Thus the Lindeberg condition holds for every fixed linear combination of the centered scores.

Since

$$\mathrm{Var}_n(\psi_i) \longrightarrow \Sigma_{\mathrm{miss}},$$

the Lindeberg–Feller theorem and the Cramér–Wold device give

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \{\psi_i - \mathbb{E}_n(\psi_i)\} \xrightarrow{\mathrm{d}} \mathcal{N}(0, \Sigma_{\mathrm{miss}}).$$

This conclusion remains valid when Σ_{miss} is singular: any scalar projection with zero limiting variance converges to zero by Chebyshev's inequality.

Combining this result with (C.34) yields

$$\sqrt{n}\, \widehat{g}(\beta) \xrightarrow{\mathrm{d}} \mathcal{N}(\tau, \Sigma_{\mathrm{miss}}).$$

Finally, (C.32), (C.33), and Slutsky's theorem imply

$$\sqrt{n}\{\widehat{\beta}(M) - \beta\} \xrightarrow{\mathrm{d}} \mathcal{N}\left(K_M\tau,\, K_M \Sigma_{\mathrm{miss}} K_M^\top\right) = \mathcal{N}\{K_M\tau, V_{\mathrm{miss}}(M)\}.$$

By Assumption 4 (iii), $V_{\mathrm{miss}}(M) \in \mathbb{S}^p_{++}$, so the limiting distribution is nondegenerate.

The same argument applies to either single-source system whenever the corresponding block identifies the full coefficient. Its limiting mean is $Q_{\mathrm{IV}}^{-1}\tau_z$ or $Q_{\mathrm{SV}}^{-1}\tau_w$, respectively, and its limiting covariance is

$$Q_{\mathrm{IV}}^{-1}\Sigma_{\mathrm{IV,miss}}Q_{\mathrm{IV}}^{-T} \quad \text{or} \quad Q_{\mathrm{SV}}^{-1}\Sigma_{\mathrm{SV,miss}}Q_{\mathrm{SV}}^{-T}.$$

These Gaussian limits may be degenerate. Nondegenerate single-source inference additionally requires the corresponding limiting covariance to be positive definite.

□

C.6 Proof of Lemma 6

Proof of Lemma 6. Fix K satisfying $KJ = I_p$. The first-order risk is $\mathrm{AMSE}_\tau(K) = \|K\tau\|_2^2 + \mathrm{tr}(K\Sigma_{\mathrm{miss}}K^\top)$. Because the limiting covariance is common to the local directions under consideration,

$$\begin{aligned} R(K) &= \sup_{\|u\|_2 \le 1} \|KDu\|_2^2 + \mathrm{tr}(K\Sigma_{\mathrm{miss}}K^\top) \\ &= \|KD\|_{\mathrm{op}}^2 + \mathrm{tr}(K\Sigma_{\mathrm{miss}}K^\top). \end{aligned}$$

The last equality follows from the definition of the operator norm and remains valid when D is singular.

For an attainable direction, suppose the estimator's root-n error converges to $\mathcal{N}(K\tau, K\Sigma_{\mathrm{miss}}K^\top)$. Uniform integrability of its squared root-n error implies convergence of the corresponding expectations. Thus $\lim_n n\mathbb{E}_\tau\|\widehat{\beta} - \beta\|_2^2 = \|K\tau\|_2^2 + \mathrm{tr}(K\Sigma_{\mathrm{miss}}K^\top)$. This establishes the connection with actual AMSE.

The supremum is taken after defining the pointwise asymptotic risk. No interchange of a supremum and a limit is used. If some directions are unattainable, taking the supremum over the larger set supplies the stated conservative envelope. □

C.7 Proof of Proposition 7

Proof. By Theorem 5 and consistency of the sample design moments,

$$\widehat{\beta}(M_0) - \beta = o_p(1), \qquad \widehat{J} - J = o_p(1).$$

The estimated projection coefficients are consistent, and the sample disturbance cross moments converge to their row-specific population values. Replacing the disturbances by the preliminary residuals preserves this convergence. For example,

$$\frac{1}{n}Z^\top\{Y - X\widehat{\beta}(M_0)\} - \mathbb{E}_n(z_i\varsigma_i) = \left\{\frac{1}{n}Z^\top\varsigma - \mathbb{E}_n(z_i\varsigma_i)\right\} - \frac{1}{n}Z^\top X\{\widehat{\beta}(M_0) - \beta\} = o_p(1).$$

The same argument applies to the cross moments involving x_i and w_i. Moreover, Cauchy–Schwarz gives

$$\sup_n \|\mathbb{E}_n(z_i\varsigma_i)\|_2 \le \{\mathbb{E}\|z_i\|_2^2\}^{1/2}\{\sup_n \mathbb{E}_n(\varsigma_i^2)\}^{1/2} < \infty,$$

with the analogous bounds for x_i and w_i.

Expanding the score formulas in (3.5), the entries of $\widehat{\psi}_i - \psi_i$ are finite sums of entries of

$$x_ix_i^\top, \quad x_iz_i^\top, \quad x_iw_i^\top, \quad z_iz_i^\top, \quad w_iw_i^\top, \quad x_i\varsigma_i, \quad z_i\varsigma_i, \quad w_i\varsigma_i,$$

multiplied by coefficient differences of order $o_p(1)$. The moment conditions imply

$$\frac{1}{n}\sum_{i=1}^n\big\{\|x_ix_i^\top\|_{\mathrm{op}}^2 + \|x_iz_i^\top\|_{\mathrm{op}}^2 + \|x_iw_i^\top\|_{\mathrm{op}}^2 + \|z_iz_i^\top\|_{\mathrm{op}}^2 + \|w_iw_i^\top\|_{\mathrm{op}}^2$$

$$+ \|x_i\varsigma_i\|_2^2 + \|z_i\varsigma_i\|_2^2 + \|w_i\varsigma_i\|_2^2\big\} = O_p(1).$$

For the last two terms, the required bounds follow from the identities

$$z_i\varsigma_i = z_i\{\varsigma_i - z_i^\top\Sigma_z^{-1}\mathbb{E}_n(z_i\varsigma_i)\} + z_iz_i^\top\Sigma_z^{-1}\mathbb{E}_n(z_i\varsigma_i),$$

$$w_i\varsigma_i = w_i\{\varsigma_i - w_i^\top\Sigma_w^{-1}\mathbb{E}_n(w_i\varsigma_i)\} + w_iw_i^\top\Sigma_w^{-1}\mathbb{E}_n(w_i\varsigma_i).$$

Thus, since the dimensions are fixed,

$$\frac{1}{n}\sum_{i=1}^n\|\widehat{\psi}_i - \psi_i\|_2^2 = o_p(1).$$

The uniform $2+\eta$ score-moment bound implies that each entry of

$$\{\psi_i - \mathbb{E}_n(\psi_i)\}\{\psi_i - \mathbb{E}_n(\psi_i)\}^\top$$

has a uniformly bounded $1+\eta/2$ moment. Since observations are independent within each row and the dimension is fixed, the triangular-array weak law of large numbers gives

$$\frac{1}{n}\sum_{i=1}^{n}\{\psi_i - \mathbb{E}_n(\psi_i)\}\{\psi_i - \mathbb{E}_n(\psi_i)\}^\top = \mathrm{Var}_n(\psi_i) + o_p(1) = \Sigma_{\mathrm{miss}} + o_p(1).$$

Also,

$$\frac{1}{n}\sum_{i=1}^{n}\psi_i - \mathbb{E}_n(\psi_i) = O_p(n^{-1/2}),$$

so sample centering changes the preceding matrix by $o_p(1)$. By Cauchy–Schwarz,

$$\frac{1}{n}\sum_{i=1}^{n}\|\widehat{\psi}_i - \psi_i\|_2\|\psi_i\|_2 \le \left\{\frac{1}{n}\sum_{i=1}^{n}\|\widehat{\psi}_i - \psi_i\|_2^2\right\}^{1/2}\left\{\frac{1}{n}\sum_{i=1}^{n}\|\psi_i\|_2^2\right\}^{1/2} = o_p(1).$$

Consequently,

$$\widehat{\Sigma} - \Sigma_{\mathrm{miss}} = o_p(1).$$

We next establish consistency of the optimizer. Define

$$R(K) = \|KD\|_{\mathrm{op}}^2 + \mathrm{tr}(K\Sigma_{\mathrm{miss}}K^\top), \qquad \widehat{R}(K) = \|KD\|_{\mathrm{op}}^2 + \mathrm{tr}(K\widehat{\Sigma}K^\top).$$

Since $\Sigma_{\mathrm{miss}} \succ 0$ and $\mathrm{rank}(J) = p$, the population problem has a unique minimizer. The sample problem also has a unique minimizer with probability approaching one, because

$$\Pr\{\mathrm{rank}(\widehat{J}) = p,\ \widehat{\Sigma} \succ 0\} \longrightarrow 1.$$

On this event, all sample optimizers below are well defined. On its complement, assign the estimator any fixed value; this convention does not affect the stated convergence in probability or distribution.

To compare the two constraint sets, let

$$\widetilde{K} = K^* + (I_p - K^* \widehat{J})(\widehat{J}^\top \widehat{J})^{-1} \widehat{J}^\top.$$

Then

$$\widetilde{K}\widehat{J} = I_p, \qquad \widetilde{K} - K^* = o_p(1).$$

Sample optimality therefore gives

$$\lambda_{\min}(\widehat{\Sigma}) \|\widehat{K}^*\|_{\mathrm{F}}^2 \le \widehat{R}(\widehat{K}^*) \le \widehat{R}(\widetilde{K}) = R(K^*) + o_p(1).$$

Since $\lambda_{\min}(\widehat{\Sigma}) \xrightarrow{\mathrm{p}} \lambda_{\min}(\Sigma_{\mathrm{miss}}) > 0$, it follows that

$$\|\widehat{K}^*\|_{\mathrm{F}} = O_p(1),$$

and hence

$$\left| \widehat{R}(\widehat{K}^*) - R(\widehat{K}^*) \right| \le \|\widehat{\Sigma} - \Sigma_{\mathrm{miss}}\|_{\mathrm{op}} \|\widehat{K}^*\|_{\mathrm{F}}^2 = o_p(1).$$

Now let

$$\overline{K} = \widehat{K}^* + (I_p - \widehat{K}^* J)(J^\top J)^{-1} J^\top.$$

Using $\widehat{K}^* \widehat{J} = I_p$, we obtain

$$\overline{K} J = I_p, \qquad I_p - \widehat{K}^* J = \widehat{K}^* (\widehat{J} - J) = o_p(1).$$

Thus $\overline{K} - \widehat{K}^* = o_p(1)$, and boundedness of $\widehat{K}^*$ gives

$$R(\overline{K}) - R(\widehat{K}^*) = o_p(1).$$

Combining these relations with sample optimality yields

$$\begin{aligned} 0 &\le R(\overline{K}) - R(K^*) \\ &= \widehat{R}(\widehat{K}^*) - R(K^*) + o_p(1) \\ &\le \widehat{R}(\widetilde{K}) - R(K^*) + o_p(1) = o_p(1). \end{aligned}$$

Strong convexity on $\{K : KJ = I_p\}$ then implies

$$\lambda_{\min}(\Sigma_{\text{miss}})\|\overline{K} - K^*\|_{\text{F}}^2 \le R(\overline{K}) - R(K^*) = o_p(1).$$

Therefore,

$$\widehat{K}^* - K^* = (\widehat{K}^* - \overline{K}) + (\overline{K} - K^*) = o_p(1).$$

Finally, the sample constraint gives the exact identity

$$\sqrt{n}(\widehat{\beta}_{\text{opt}} - \beta) = \widehat{K}^* \sqrt{n}\, \widehat{g}(\beta).$$

Theorem 5 yields $\sqrt{n}\, \widehat{g}(\beta) \xrightarrow{\text{d}} \mathcal{N}(\tau, \Sigma_{\text{miss}})$. By Slutsky's theorem,

$$\sqrt{n}(\widehat{\beta}_{\text{opt}} - \beta) \xrightarrow{\text{d}} \mathcal{N}\left(K^*\tau,\, K^*\Sigma_{\text{miss}}K^{*\top}\right).$$

Under exact specification with $D = 0$ and $\Sigma_{\text{miss}} = \Sigma$,

$$K^* = (J^\top \Sigma^{-1} J)^{-1} J^\top \Sigma^{-1}, \qquad K^* \Sigma K^{*\top} = (J^\top \Sigma^{-1} J)^{-1}.$$

Since $\tau = 0$, the preceding limit reduces to

$$\sqrt{n}(\widehat{\beta}_{\text{opt}} - \beta) \xrightarrow{\text{d}} \mathcal{N}\left(0, (J^\top \Sigma^{-1} J)^{-1}\right).$$

□

C.8 Proof of Proposition 8

Proof of Proposition 8. Let $K^* = (J^\top \Sigma^{-1} J)^{-1} J^\top \Sigma^{-1}$. Then $K^* J = I_p$ and $K^* \Sigma K^{*\top} = (J^\top \Sigma^{-1} J)^{-1}$.

For any feasible K, define $L := K - K^*$, the difference between the two sensitivities. Since both sensitivities are left inverses of J, $LJ = 0_{p\times p}$. Moreover, $K^*\Sigma = (J^\top \Sigma^{-1} J)^{-1} J^\top$, so $K^* \Sigma L^\top = 0_{p\times p}$. The covariance expansion therefore reduces to

$$K\Sigma K^\top = K^*\Sigma K^{*\top} + L\Sigma L^\top.$$

The added term is positive semidefinite. Hence K^* minimizes the covariance in the Loewner order and, in particular, its trace.

The single-source sensitivities $K^{(\text{IV})} = (Q_{\text{IV}}^{-1}, 0_{p\times p})$ and $K^{(\text{SV})} = (0_{p\times p}, Q_{\text{SV}}^{-1})$ satisfy $K^{(\text{IV})} J = K^{(\text{SV})} J = I_p$. Their covariance traces are $\text{tr}(Q_{\text{IV}}^{-1} \Sigma_{\text{IV}} Q_{\text{IV}}^{-T})$ and $\text{tr}(Q_{\text{SV}}^{-1} \Sigma_{\text{SV}} Q_{\text{SV}}^{-T})$. Applying the preceding comparison proves the claimed inequality.

The exact-specification limits have mean zero. If the corresponding squared root-n errors are uniformly integrable, their expectations converge to the traces of the limiting covariances, establishing the AMSE comparison. □

C.9 Proof of Proposition 9

Proof of Proposition 9. The common criterion is $R(K) = \|KD\|_{\mathrm{op}}^2 + \mathrm{tr}(K\Sigma_{\mathrm{miss}}K^\top)$. Positive definiteness of Σ_{miss} makes this objective continuous, strongly convex, and coercive. The feasible set $\{K : KJ = I_p\}$ is closed and nonempty because J has full column rank. Thus a unique minimizer K^* exists.

The identified single-source sensitivities satisfy $K^{(\mathrm{IV})}J = Q_{\mathrm{IV}}^{-1}Q_{\mathrm{IV}} = I_p$ and $K^{(\mathrm{SV})}J = Q_{\mathrm{SV}}^{-1}Q_{\mathrm{SV}} = I_p$. They are therefore feasible competitors, and optimality gives $R(K^*) \le R\{K^{(\mathrm{IV})}\}$ and $R(K^*) \le R\{K^{(\mathrm{SV})}\}$.

Lemma 6 identifies these criteria with worst-case local AMSE when all directions are attainable and the required asymptotic distributions and uniform-integrability conditions hold. Otherwise, the inequalities compare the first-order envelopes. Neither conclusion requires, or implies, dominance at every individual local direction. □

C.10 Proof of Proposition 10

Proof of Proposition 10. Use the sample information matrices $S_Z = X^\top P_Z X$ and $S_W = X^\top P_W^\perp X$. Then $M_\lambda = n\,\mathrm{diag}\{(1-\lambda)S_Z^{-1}, \lambda S_W^{-1}\}$, while $\widehat{J} = n^{-1}(S_Z^\top, S_W^\top)^\top$. Since both information matrices are symmetric,

$$\widehat{J}^\top M_\lambda = \big((1-\lambda)I_p, \lambda I_p\big), \qquad \widehat{J}^\top M_\lambda \widehat{J} = \frac{1}{n}\{(1-\lambda)S_Z + \lambda S_W\}.$$

The second matrix is positive definite for every $\lambda \in [0,1]$, because S_Z and S_W are positive definite.

The GMM first-order condition is therefore $(1-\lambda)X^\top P_Z(Y-Xb)+\lambda X^\top P_W^\perp(Y-Xb)=0_p$. This is also the first-order condition of the WLS criterion. Both criteria are strictly convex in b, so they have the same unique minimizer: $\widehat{\beta}(M_\lambda)=\widehat{\beta}_\lambda$.

For $\lambda\in(0,1)$, both diagonal blocks of M_λ are positive definite. At the endpoints, one block is zero and the weight is positive semidefinite, but the parameter Hessian remains positive definite. The same calculation therefore recovers the IV-only and SV-only estimators. □

D. Raw versus projected IV information

The projected representation also places the IV and SV restrictions on the same p-dimensional scale. The next result shows that this compression does not sacrifice any IV identifying direction.

Proposition 11 (Projection preserves identifying directions)**.** Under (2.1), suppose the following conditions hold.

(i) The instrument covariance matrix satisfies $\Sigma_z\in\mathbb{S}^q_{++}$.

(ii) The projected IV and residualized SV moments are well defined, with $Q_{\text{IV}}=\Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$ and $Q_{\text{SV}}=\mathbb{E}\{(\Pi_w^\perp x_i)x_i^\top\}$.

Then $\ker(Q_{\text{IV}})=\ker(\Sigma_{zx})$. Consequently, for $J=\{Q_{\text{IV}}^\top,Q_{\text{SV}}^\top\}^\top$ and $J_{\text{raw}}:=\{\Sigma_{zx}^\top,Q_{\text{SV}}^\top\}^\top$, we have $\ker(J)=\ker(J_{\text{raw}})$ and $\operatorname{rank}(J)=\operatorname{rank}(J_{\text{raw}})$. Hence, the projected-IV-plus-residualized-SV system and the corresponding raw-IV-plus-residualized-SV system preserve exactly the same identifying directions for β, and $\operatorname{rank}(J)=p$ if and only if $\operatorname{rank}(J_{\text{raw}})=p$.

Moreover, if $\operatorname{rank}(\Sigma_{zx}) = p$, any linear compression of the q-dimensional IV moment that preserves all IV identifying directions must have dimension at least p. The projected instrument $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i$ has dimension p and therefore attains this minimum.

The main analysis adopts the projected IV moment because it places the IV and SV channels on a common p-dimensional scale while preserving all IV identifying directions. Projection, however, need not preserve all efficiency information contained in the original q-dimensional IV moment. This section isolates that distinction by comparing the raw and projected IV systems directly, without involving the SV block.

Recall that the projected IV restriction is $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = 0_p$, whereas the classical raw-IV restriction is $\mathbb{E}(z_i\varsigma_i) = 0_q$. Since $\mathbb{E}\{(\Pi_z x_i)\varsigma_i\} = \Sigma_{xz}\Sigma_z^{-1}\mathbb{E}(z_i\varsigma_i)$, classical IV validity implies projected IV validity. The converse need not hold when $q > p$, because components of $\mathbb{E}(z_i\varsigma_i)$ lying in $\ker(\Sigma_{xz}\Sigma_z^{-1})$ are removed by projection and therefore do not affect the projected restriction. Thus, the comparison between the two representations involves a validity–efficiency trade-off. The raw IV system retains all q moment restrictions and may exploit additional overidentifying information, but requires the stronger condition that all of these directions are valid. The projected system requires validity only after compression to the component relevant for predicting x_i, at the possible cost of discarding useful overidentifying information when the stronger classical restriction is in fact correct.

For the remainder of this section, suppose that $\mathbb{E}(z_i\varsigma_i) = 0_q$. Define the raw IV score by $\psi^{\mathrm{IV}}_{\mathrm{raw},i} := z_i\varsigma_i \in \mathbb{R}^q$, with $\Sigma^{\mathrm{IV}}_{\mathrm{raw}} := \operatorname{Var}(\psi^{\mathrm{IV}}_{\mathrm{raw},i})$, and define $T_{\mathrm{IV}} := \Sigma_{xz}\Sigma_z^{-1} \in \mathbb{R}^{p\times q}$. Under

classical IV validity, the projected IV score satisfies $\psi_{\text{IV},i} = T_{\text{IV}}\psi^{\text{IV}}_{\text{raw},i}$, so that $\Sigma_{\text{IV}} = T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^{\top}$. The corresponding Jacobians are Σ_{zx} for the raw IV system and $Q_{\text{IV}} = T_{\text{IV}}\Sigma_{zx}$ for the projected system.

Let $\mathcal{I}^{\text{IV}}_{\text{raw}} := \Sigma_{zx}^{\top}(\Sigma^{\text{IV}}_{\text{raw}})^{-1}\Sigma_{zx}$ and $\mathcal{I}^{\text{IV}}_{\text{proj}} := Q_{\text{IV}}^{\top}\Sigma_{\text{IV}}^{-1}Q_{\text{IV}}$ denote the efficient information matrices in the raw and projected IV systems, respectively. For the equality characterization, define

$$P_{T_{\text{IV}}} := (\Sigma^{\text{IV}}_{\text{raw}})^{1/2}T_{\text{IV}}^{\top}(T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^{\top})^{-1}T_{\text{IV}}(\Sigma^{\text{IV}}_{\text{raw}})^{1/2}$$

.

Proposition 12 (Raw IV information dominates projected IV information)**.** Suppose the following conditions hold.

(i) The classical IV restriction is correctly specified, $\mathbb{E}(z_i\varsigma_i) = 0_q$.

(ii) The projected IV block is individually identified, $Q_{\text{IV}} \in \mathbb{S}^p_{++}$, and the projection is well defined with $\Sigma_z \in \mathbb{S}^q_{++}$.

(iii) The regularity conditions required for the first-order expansions of the raw and projected IV systems hold, and $\Sigma^{\text{IV}}_{\text{raw}} \in \mathbb{S}^q_{++}$.

Then

$$\mathcal{I}^{\text{IV}}_{\text{proj}} \preceq \mathcal{I}^{\text{IV}}_{\text{raw}}, \qquad V^{\star}_{\text{raw,IV}} := (\mathcal{I}^{\text{IV}}_{\text{raw}})^{-1} \preceq (\mathcal{I}^{\text{IV}}_{\text{proj}})^{-1} =: V^{\star}_{\text{proj,IV}}. \tag{D.1}$$

More precisely,

$$\mathcal{I}^{\text{IV}}_{\text{raw}} - \mathcal{I}^{\text{IV}}_{\text{proj}} = \Sigma_{zx}^{\top}(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}(I_q - P_{T_{\text{IV}}})(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}\Sigma_{zx} \succeq 0. \tag{D.2}$$

Hence, projection is information-lossless if and only if $\mathcal{C}\{(\Sigma_{\text{raw}}^{\text{IV}})^{-1}\Sigma_{zx}\} \subseteq \mathcal{C}(T_{\text{IV}}^{\top})$.

Proposition 12 separates preservation of identification from preservation of efficiency. By Proposition 11, the projected IV moment preserves exactly the same identifying directions as the raw IV moment. Hence, the additional directions available when $q > p$ are not required for point identification of β. They may nevertheless contain useful overidentifying information. When the stronger classical restriction $\mathbb{E}(z_i\varsigma_i) = 0_q$ is valid, retaining these directions can reduce asymptotic variance, whereas projection may discard part of this information.

The equality condition has a direct geometric interpretation. The space $\mathcal{C}(T_{\text{IV}}^{\top})$ contains the raw-IV directions retained by the projection, whereas $\mathcal{C}\{(\Sigma_{\text{raw}}^{\text{IV}})^{-1}\Sigma_{zx}\}$ contains the efficient raw-IV directions relevant for estimating β. Projection is therefore information-lossless precisely when every efficient raw-IV direction is already contained in the subspace retained by T_{IV}. If some efficient direction lies outside that subspace, the projected system still preserves identification but loses information that could otherwise improve precision.

This result clarifies the trade-off underlying the projected formulation. The raw IV system can be more efficient when all q classical IV moment restrictions are valid, but this gain relies on the stronger validity requirement. The projected formulation instead requires only the weaker projected restriction in the main analysis and provides an identification-preserving p-dimensional representation that can be placed on the same scale as the surrogate-control moment. Its role is therefore not to dominate the raw IV system in efficiency, but to provide the common representation required for the unified IV–SV analysis.

When $q = p$ and $Q_{\text{IV}} \in \mathbb{S}^p_{++}$, T_{IV} is nonsingular, so the raw and projected IV moments are related by an invertible transformation. Consequently, $\mathcal{I}^{\text{IV}}_{\text{proj}} = \mathcal{I}^{\text{IV}}_{\text{raw}}$, and no IV information is lost through projection. The validity–efficiency trade-off is therefore relevant primarily in the overidentified case $q > p$.

D.1 Proof of Proposition 11

Proof of Proposition 11. We first show that the projected IV Jacobian and the raw IV Jacobian have the same null space. Recall that $Q_{\text{IV}} = \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx}$. For any $a \in \mathbb{R}^p$,

$$a^\top Q_{\text{IV}} a = a^\top \Sigma_{xz}\Sigma_z^{-1}\Sigma_{zx} a = \left\| \Sigma_z^{-1/2}\Sigma_{zx} a \right\|_2^2 .$$

Since $\Sigma_z \in \mathbb{S}^q_{++}$, the right-hand side is zero if and only if $\Sigma_{zx} a = 0_q$. Moreover, $Q_{\text{IV}} \succeq 0$, so $Q_{\text{IV}} a = 0_p$ if and only if $a^\top Q_{\text{IV}} a = 0$. Therefore,

$$\ker(Q_{\text{IV}}) = \ker(\Sigma_{zx}). \tag{D.3}$$

We next compare the identifying directions of the projected and raw combined systems. By definition,

$$J = \begin{pmatrix} Q_{\text{IV}} \\ \\ Q_{\text{SV}} \end{pmatrix}, \qquad J_{\text{raw}} = \begin{pmatrix} \Sigma_{zx} \\ \\ Q_{\text{SV}} \end{pmatrix}.$$

For any $a \in \mathbb{R}^p$, $Ja = 0_{2p}$ if and only if $Q_{\text{IV}} a = 0_p$ and $Q_{\text{SV}} a = 0_p$. By (D.3), this is equivalent to $\Sigma_{zx} a = 0_q$ and $Q_{\text{SV}} a = 0_p$, which in turn is equivalent to $J_{\text{raw}} a = 0_{q+p}$. Hence,

$$\ker(J) = \ker(J_{\text{raw}}).$$

Because both matrices have p columns, the rank–nullity theorem then gives

$$\operatorname{rank}(J) = \operatorname{rank}(J_{\text{raw}}).$$

Thus, replacing the raw IV block by the projected IV block does not remove any parameter direction that can be identified by the combined IV–SV system. In particular, $\operatorname{rank}(J) = p$ if and only if $\operatorname{rank}(J_{\text{raw}}) = p$.

It remains to establish the minimal dimension of an identification-preserving linear IV compression. Suppose $\operatorname{rank}(\Sigma_{zx}) = p$, and consider any k-dimensional linear compression of the raw q-dimensional IV moment. The Jacobian of the resulting compressed moment has rank at most k. Therefore, preserving all p identifying directions of Σ_{zx} necessarily requires $k \geq p$.

Finally, $\Pi_z x_i = \Sigma_{xz}\Sigma_z^{-1} z_i \in \mathbb{R}^p$, so the projected IV moment has exactly p components. By (D.3), its Jacobian Q_{IV} has the same null space, and hence the same identifying directions, as Σ_{zx}. Therefore, the projected IV block preserves all IV identifying directions using the smallest possible number p of linearly compressed moments.

This completes the proof.

□

D.2 Proof of Proposition 12

Proof of Proposition 12. Recall that $T_{\text{IV}} = \Sigma_{xz}\Sigma_z^{-1}$, $\Sigma_{\text{IV}} = T_{\text{IV}}\Sigma_{\text{raw}}^{\text{IV}}T_{\text{IV}}^\top$, and $Q_{\text{IV}} = T_{\text{IV}}\Sigma_{zx}$. The corresponding efficient information matrices are $\mathcal{I}_{\text{raw}}^{\text{IV}} = \Sigma_{zx}^\top(\Sigma_{\text{raw}}^{\text{IV}})^{-1}\Sigma_{zx}$ and $\mathcal{I}_{\text{proj}}^{\text{IV}} = Q_{\text{IV}}^\top\Sigma_{\text{IV}}^{-1}Q_{\text{IV}}$.

We first verify that Σ_{IV} is nonsingular. Since $Q_{\text{IV}} = T_{\text{IV}}\Sigma_{zx} \in \mathbb{S}^p_{++}$, the matrix $T_{\text{IV}} \in \mathbb{R}^{p\times q}$ has full row rank p. Together with $\Sigma^{\text{IV}}_{\text{raw}} \in \mathbb{S}^q_{++}$, this implies $\Sigma_{\text{IV}} = T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^\top \in \mathbb{S}^p_{++}$. Hence, Σ_{IV}^{-1} and $P_{T_{\text{IV}}}$ are well defined.

Using $Q_{\text{IV}} = T_{\text{IV}}\Sigma_{zx}$, we can write

$$\begin{aligned}\mathcal{I}^{\text{IV}}_{\text{proj}} &= \Sigma_{zx}^\top T_{\text{IV}}^\top \left(T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^\top\right)^{-1} T_{\text{IV}}\Sigma_{zx} \\ &= \Sigma_{zx}^\top (\Sigma^{\text{IV}}_{\text{raw}})^{-1/2} P_{T_{\text{IV}}} (\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}\Sigma_{zx}, \end{aligned} \tag{D.4}$$

where $P_{T_{\text{IV}}} = (\Sigma^{\text{IV}}_{\text{raw}})^{1/2}T_{\text{IV}}^\top(T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^\top)^{-1}T_{\text{IV}}(\Sigma^{\text{IV}}_{\text{raw}})^{1/2}$.

The matrix $P_{T_{\text{IV}}}$ is symmetric. Moreover, $P^2_{T_{\text{IV}}} = P_{T_{\text{IV}}}$, since the middle product $T_{\text{IV}}\Sigma^{\text{IV}}_{\text{raw}}T_{\text{IV}}^\top$ cancels with its inverse. Hence, $P_{T_{\text{IV}}}$ is the orthogonal projector onto $\mathcal{C}\{(\Sigma^{\text{IV}}_{\text{raw}})^{1/2}T_{\text{IV}}^\top\}$. It follows that $I_q - P_{T_{\text{IV}}} \succeq 0$.

The raw information matrix can similarly be written as $\mathcal{I}^{\text{IV}}_{\text{raw}} = \Sigma_{zx}^\top(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}I_q(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}\Sigma_{zx}$. Subtracting (D.4) therefore yields

$$\mathcal{I}^{\text{IV}}_{\text{raw}} - \mathcal{I}^{\text{IV}}_{\text{proj}} = \Sigma_{zx}^\top(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}(I_q - P_{T_{\text{IV}}})(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}\Sigma_{zx} \succeq 0. \tag{D.5}$$

Thus, $\mathcal{I}^{\text{IV}}_{\text{proj}} \preceq \mathcal{I}^{\text{IV}}_{\text{raw}}$.

We next translate the information ordering into an asymptotic covariance ordering. Since $Q_{\text{IV}} \in \mathbb{S}^p_{++}$, the projected Jacobian has full rank p. By $\ker(Q_{\text{IV}}) = \ker(\Sigma_{zx})$ from Proposition 11, Σ_{zx} also has full column rank p. Together with positive definiteness of $\Sigma^{\text{IV}}_{\text{raw}}$ and Σ_{IV}, this gives $\mathcal{I}^{\text{IV}}_{\text{raw}}, \mathcal{I}^{\text{IV}}_{\text{proj}} \in \mathbb{S}^p_{++}$. Since inversion reverses the Loewner order on positive definite matrices,

$$V^\star_{\text{raw,IV}} = (\mathcal{I}^{\text{IV}}_{\text{raw}})^{-1} \preceq (\mathcal{I}^{\text{IV}}_{\text{proj}})^{-1} = V^\star_{\text{proj,IV}}.$$

It remains to characterize when projection is information-lossless. By (D.5), $\mathcal{I}^{\text{IV}}_{\text{raw}} = \mathcal{I}^{\text{IV}}_{\text{proj}}$ if and only if

$$(I_q - P_{T_{\text{IV}}})(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}\Sigma_{zx} = 0.$$

Because $P_{T_{\text{IV}}}$ projects onto $\mathcal{C}\{(\Sigma^{\text{IV}}_{\text{raw}})^{1/2}T_{\text{IV}}^{\top}\}$, this is equivalent to

$$\mathcal{C}\left\{(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}\Sigma_{zx}\right\} \subseteq \mathcal{C}\left\{(\Sigma^{\text{IV}}_{\text{raw}})^{1/2}T_{\text{IV}}^{\top}\right\}.$$

Applying the nonsingular transformation $(\Sigma^{\text{IV}}_{\text{raw}})^{-1/2}$ to both spaces gives the equivalent condition

$$\mathcal{C}\left\{(\Sigma^{\text{IV}}_{\text{raw}})^{-1}\Sigma_{zx}\right\} \subseteq \mathcal{C}(T_{\text{IV}}^{\top}).$$

Hence, projection is information-lossless exactly when every efficient raw-IV direction relevant for estimating β lies in the raw-IV subspace retained by T_{IV}. Otherwise, $I_q - P_{T_{\text{IV}}}$ removes a nonzero component of at least one such direction, so the projected system preserves identification but loses efficiency information.

This completes the proof.

□

E. Conditional risk along the restricted WLS path

This appendix studies the WLS path in Section 4.4 conditionally on the realized design matrices. The analysis characterizes finite-sample conditional squared-error risk and does not impose conditional validity of the projected moments.

Recall $S_Z := X^\top P_Z X$, $S_W := X^\top P_W^\perp X$, $S_\lambda := (1-\lambda)S_Z + \lambda S_W$, $\mathcal{D} := \sigma(X, Z, W)$, $U_Z := X^\top P_Z \varsigma$, and $U_W := X^\top P_W^\perp \varsigma$. Define $c_Z := \mathbb{E}(\|U_Z\|_2^2 \mid \mathcal{D})$, $c_{ZW} := \mathbb{E}(U_Z^\top U_W \mid \mathcal{D})$, and $c_W := \mathbb{E}(\|U_W\|_2^2 \mid \mathcal{D})$. These quantities are conditional second moments rather than conditional variances.

Proposition 13 (Conditional risk bounds and a tractable surrogate)**.** Condition on a realized design for which $S_Z, S_W \in \mathbb{S}_{++}^p$ and $c_Z, c_W < \infty$. Then, for every $\lambda \in [0,1]$, $\widehat{\beta}_\lambda - \beta = S_\lambda^{-1}\{(1-\lambda)U_Z + \lambda U_W\}$. The conditional risk in (4.6) satisfies

$$\frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\lambda_{\max}(S_\lambda)^2} \le \mathcal{R}_{\mathrm{tr}}(\lambda) \le \frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\lambda_{\min}(S_\lambda)^2}. \quad \text{(E.1)}$$

Define the surrogate criterion

$$\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) := \frac{(1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W}{\{(1-\lambda)\lambda_{\min}(S_Z) + \lambda\lambda_{\min}(S_W)\}^2}. \quad \text{(E.2)}$$

It always satisfies $\mathcal{R}_{\mathrm{tr}}(\lambda) \le \widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda)$. If, for a specified constant $C \ge 1$, $\lambda_{\max}(S_Z) \le C\lambda_{\min}(S_Z)$ and $\lambda_{\max}(S_W) \le C\lambda_{\min}(S_W)$, then $C^{-2}\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) \le \mathcal{R}_{\mathrm{tr}}(\lambda) \le \widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda)$ uniformly over $\lambda \in [0,1]$. For a sequence of designs, this comparison is uniform in sample size only if the same C applies throughout the sequence.

If $(c_W - c_{ZW})\lambda_{\min}(S_Z) + (c_Z - c_{ZW})\lambda_{\min}(S_W) \neq 0$, the surrogate has at most one interior stationary point:

$$\lambda_{\mathrm{sur}}^\star = \frac{c_Z\lambda_{\min}(S_W) - c_{ZW}\lambda_{\min}(S_Z)}{(c_W - c_{ZW})\lambda_{\min}(S_Z) + (c_Z - c_{ZW})\lambda_{\min}(S_W)}. \quad \text{(E.3)}$$

Its minimum over $[0,1]$ is obtained by comparing the two endpoints and $\lambda_{\mathrm{sur}}^\star$, when the latter belongs to $(0,1)$. If the displayed denominator is zero, the derivative has a constant-sign

numerator or vanishes identically; an endpoint therefore attains the minimum.

The conditional risk also admits the decomposition $\mathcal{R}_{\mathrm{tr}}(\lambda) = \|S_\lambda^{-1}\mathbb{E}(U_\lambda \mid \mathcal{D})\|_2^2 + \mathrm{tr}\{S_\lambda^{-1}\,\mathrm{Var}(U_\lambda \mid \mathcal{D})S_\lambda^{-1}\}$, where $U_\lambda = (1-\lambda)U_Z + \lambda U_W$. Consequently, unconditional IV or SV validity does not justify dropping the conditional bias term.

Whenever the conditional covariance of ς exists,

$$c_Z = \mathrm{tr}\{X^\top P_Z\,\mathrm{Var}(\varsigma \mid \mathcal{D})P_Z X\} + \|\mathbb{E}(U_Z \mid \mathcal{D})\|_2^2,$$

$$c_{ZW} = \mathrm{tr}\{X^\top P_Z\,\mathrm{Var}(\varsigma \mid \mathcal{D})P_W^\perp X\} + \mathbb{E}(U_Z \mid \mathcal{D})^\top \mathbb{E}(U_W \mid \mathcal{D}),$$

$$c_W = \mathrm{tr}\{X^\top P_W^\perp\,\mathrm{Var}(\varsigma \mid \mathcal{D})P_W^\perp X\} + \|\mathbb{E}(U_W \mid \mathcal{D})\|_2^2.$$

Thus the surrogate balances conditional second moments against lower bounds on information strength. It is an upper bound on the true conditional risk, with a controlled multiplicative comparison when both information matrices have bounded condition numbers.

The candidate $\lambda_{\mathrm{sur}}^\star$ minimizes the surrogate only after comparison with the endpoints; it need not minimize $\mathcal{R}_{\mathrm{tr}}(\lambda)$. Moreover, c_Z, c_{ZW}, c_W depend on unknown conditional disturbance moments, so this result describes an oracle criterion rather than a feasible tuning rule. Under the stated condition-number bound, any surrogate minimizer has conditional risk at most C^2 times the minimum conditional risk along the WLS path, although its location need not be close to the true risk minimizer.

E.1 Proof of Proposition 13

Proof of Proposition 13. Let $\mathcal{D} := \sigma(X, Z, W)$ denote the sigma-field generated by the realized design matrices. Throughout the proof, all conditional expectations and variances are taken with respect to $\mathcal{D}$.

Recall $U_Z := X^\top P_Z \varsigma$, $U_W := X^\top P_W^\perp \varsigma$, and $U_\lambda := (1-\lambda)U_Z + \lambda U_W$. Substituting $Y = X\beta + \varsigma$ into (4.5) gives $\widehat{\beta}_\lambda = S_\lambda^{-1}(S_\lambda \beta + U_\lambda)$, and hence $\widehat{\beta}_\lambda - \beta = S_\lambda^{-1} U_\lambda$.

Define $\mu_Z := \mathbb{E}(U_Z \mid \mathcal{D})$, $\mu_W := \mathbb{E}(U_W \mid \mathcal{D})$, and $\mu_\lambda := \mathbb{E}(U_\lambda \mid \mathcal{D}) = (1-\lambda)\mu_Z + \lambda \mu_W$. Since S_λ is measurable with respect to $\mathcal{D}$, $\mathbb{E}(\widehat{\beta}_\lambda - \beta \mid \mathcal{D}) = S_\lambda^{-1}\mu_\lambda$. Thus the estimator need not be conditionally centered even when the population moment restrictions are valid.

Let $\Omega_\lambda := \mathbb{E}(U_\lambda U_\lambda^\top \mid \mathcal{D})$. Because S_λ is symmetric positive definite, $\mathcal{R}_{\mathrm{tr}}(\lambda) = \mathrm{tr}(S_\lambda^{-1}\Omega_\lambda S_\lambda^{-1})$. Using $\Omega_\lambda = \mathrm{Var}(U_\lambda \mid \mathcal{D}) + \mu_\lambda \mu_\lambda^\top$ gives

$$\mathcal{R}_{\mathrm{tr}}(\lambda) = \|S_\lambda^{-1}\mu_\lambda\|_2^2 + \mathrm{tr}\{S_\lambda^{-1}\,\mathrm{Var}(U_\lambda \mid \mathcal{D}) S_\lambda^{-1}\}.$$

This establishes the conditional bias–variance decomposition.

Expanding $U_\lambda U_\lambda^\top$ and taking traces yields $\mathrm{tr}(\Omega_\lambda) = (1-\lambda)^2 c_Z + 2\lambda(1-\lambda)c_{ZW} + \lambda^2 c_W$. Write this expression as $N(\lambda)$. It is finite and nonnegative, since $N(\lambda) = \mathbb{E}(\|U_\lambda\|_2^2 \mid \mathcal{D})$; finiteness of c_{ZW} follows from conditional Cauchy–Schwarz.

For positive semidefinite matrices A and B, $\lambda_{\min}(B)\,\mathrm{tr}(A) \le \mathrm{tr}(AB) \le \lambda_{\max}(B)\,\mathrm{tr}(A)$. Indeed, conjugating $\lambda_{\min}(B) I \preceq B \preceq \lambda_{\max}(B) I$ by $A^{1/2}$ and taking traces proves these inequal-

ities. Applying them to $\mathcal{R}_{\mathrm{tr}}(\lambda) = \mathrm{tr}(\Omega_\lambda S_\lambda^{-2})$ gives

$$\frac{N(\lambda)}{\lambda_{\max}(S_\lambda)^2} \le \mathcal{R}_{\mathrm{tr}}(\lambda) \le \frac{N(\lambda)}{\lambda_{\min}(S_\lambda)^2},$$

because the extreme eigenvalues of S_λ^{-2} are $\lambda_{\max}(S_\lambda)^{-2}$ and $\lambda_{\min}(S_\lambda)^{-2}$. This proves (E.1).

We next compare the conditional risk with its surrogate. Define $a_Z := \lambda_{\min}(S_Z)$, $a_W := \lambda_{\min}(S_W)$, and $d_\lambda := (1-\lambda)a_Z + \lambda a_W$. Since S_Z, S_W are positive definite, $d_\lambda > 0$ throughout $[0,1]$. The Rayleigh-quotient characterization gives $\lambda_{\min}(S_\lambda) \ge d_\lambda$ and $\lambda_{\max}(S_\lambda) \le (1-\lambda)\lambda_{\max}(S_Z) + \lambda\lambda_{\max}(S_W)$. Consequently, $\mathcal{R}_{\mathrm{tr}}(\lambda) \le N(\lambda)/d_\lambda^2 = \widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda)$.

If the stated condition-number bounds hold with a common $C \ge 1$, then $\lambda_{\max}(S_\lambda) \le C d_\lambda$. Combining this with the preceding risk bounds gives

$$C^{-2}\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) \le \mathcal{R}_{\mathrm{tr}}(\lambda) \le \widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) \qquad \text{for all } \lambda \in [0,1].$$

For a sequence of designs, the comparison is uniform in sample size when the same C applies throughout.

It remains to characterize the stationary points of the surrogate. Let $D(\lambda) := (1-\lambda)a_Z + \lambda a_W = d_\lambda$. Then $\widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) = N(\lambda)/D(\lambda)^2$, where $N(\lambda) = c_Z + 2(c_{ZW} - c_Z)\lambda + (c_Z - 2c_{ZW} + c_W)\lambda^2$. Thus $N'(\lambda) = 2(c_{ZW} - c_Z) + 2(c_Z - 2c_{ZW} + c_W)\lambda$ and $D'(\lambda) = a_W - a_Z$. Differentiation gives $\widetilde{\mathcal{R}}'_{\mathrm{tr}}(\lambda) = \{N'(\lambda)D(\lambda) - 2N(\lambda)D'(\lambda)\}/D(\lambda)^3$. The quadratic terms in the numerator cancel, leaving

$$\widetilde{\mathcal{R}}'_{\mathrm{tr}}(\lambda) = \frac{2\big[\{(c_W - c_{ZW})a_Z + (c_Z - c_{ZW})a_W\}\lambda - (c_Z a_W - c_{ZW}a_Z)\big]}{D(\lambda)^3}.$$

If $(c_W - c_{ZW})a_Z + (c_Z - c_{ZW})a_W \neq 0$, the derivative has at most one zero, given by

$\lambda^\star_{\mathrm{sur}} = (c_Z a_W - c_{ZW} a_Z)/\{(c_W - c_{ZW})a_Z + (c_Z - c_{ZW})a_W\}$. Substituting the definitions of a_Z and a_W yields (E.3).

If instead $(c_W - c_{ZW})a_Z + (c_Z - c_{ZW})a_W = 0$, the derivative reduces to $-2(c_Z a_W - c_{ZW} a_Z)/D(\lambda)^3$. Since $D(\lambda) > 0$, it has a constant nonzero sign when $c_Z a_W - c_{ZW} a_Z \neq 0$, so the surrogate is strictly monotone and attains its minimum at an endpoint. When $c_Z a_W - c_{ZW} a_Z = 0$ as well, the derivative vanishes identically and the surrogate is constant, so either endpoint attains the minimum.

Finally, the surrogate is continuous on the compact interval $[0, 1]$ and therefore attains a minimum. In the nondegenerate case, it suffices to compare its values at the two endpoints and at $\lambda^\star_{\mathrm{sur}}$ when the latter belongs to $(0, 1)$. In the degenerate case, the preceding argument shows that an endpoint attains the minimum.

The uniform comparison also implies

$$\min_{\lambda \in [0,1]} \widetilde{\mathcal{R}}_{\mathrm{tr}}(\lambda) \leq C^2 \min_{\lambda \in [0,1]} \mathcal{R}_{\mathrm{tr}}(\lambda).$$

At any surrogate minimizer, the true conditional risk is bounded above by the surrogate minimum. Hence its conditional risk is at most C^2 times the minimum true conditional risk along the path.

This completes the proof.

□

F. Data preparation and implementation for the real-data analysis

Experimental source and batch-level profiles. We use the K562 essential-scale, day-6 Perturb-seq dataset of Replogle et al. (2022). K562 cells expressing dCas9–KRAB were transduced with a pooled lentiviral library targeting common essential genes defined by the Cancer Dependency Map. Each targeting vector expressed two single-guide RNAs directed at the same gene; non-targeting controls constituted approximately 5% of the library. GFP-positive transduced cells were enriched by fluorescence-activated cell sorting three days after infection. Six days after infection, cells were profiled using 10x Genomics Chromium single-cell RNA sequencing with direct guide capture across 48 lanes/GEM groups. These sequencing groups define the batches in our analysis.

The supplied data contain 309,915 input cells and 8,563 genes. The precomputed pseudobulk matrix groups cells by perturbation and batch and excludes groups containing fewer than five cells. Within each retained group, five cells are sampled with replacement, and their log1p expression values are averaged gene by gene. This produces 22,075 perturbation–batch profiles. The resulting observations are therefore averages of transformed expression from sampled cells, rather than sums of raw counts.

Perturbation and outcome selection. We retain perturbations represented by at least 30 pseudobulk profiles and having both batch-stratified Wilcoxon results and measured

target-gene expression. The retained data comprise 213 perturbations and 48 control profiles, totaling 8,413 profiles. Each perturbation contributes 30–48 profiles, with a median of 39. Combining each perturbation with the same 48 control profiles gives target-specific sample sizes of 78–96, with a median of 87. All target-specific analyses contain 48 batches.

For each retained perturbation, we rank the 8,563 genes by increasing Wilcoxon adjusted p-value, resolving ties by decreasing absolute z-score and then decreasing absolute log fold change. We average each gene's rank across the 213 perturbations. After excluding all retained perturbation targets, the 2,000 genes with the smallest average ranks form the common outcome panel. Ties are resolved by decreasing mean absolute z-score, decreasing mean absolute log fold change, and gene name. Every retained perturbation is analyzed against every outcome, yielding 426,000 fitted models.

Surrogate-gene selection and PCA. We exclude all retained perturbation targets and all genes appearing among the top 20 DE-ranked genes for any retained perturbation. Among the remaining genes, we select the 1,000 with the largest average DE ranks, resolving ties by increasing mean absolute z-score, increasing mean absolute log fold change, and gene name. This is an operational low-DE-response selection rule; it does not require that each selected gene be nonsignificant under every perturbation.

PCA is fitted once using these genes across all 8,413 retained perturbation and control profiles. Each gene is centered and divided by its standard deviation across profiles; features with standard deviation at most 10^{-8} are removed. We retain the first 10 principal component

scores as surrogate controls. Together, they explain approximately 3.53% of the standardized expression variance. For each target-specific model, we subset these common scores to the relevant perturbation and control profiles.

Estimation and batch-clustered inference. Within each target-specific sample, let z_i^c, x_i^c, and y_i^c denote centered perturbation indicators and expression values. Let $\widetilde{x}_i$ and $\widetilde{y}_i$ denote residuals from separate least-squares regressions of x_i and y_i on an intercept and the 10 principal component scores. The implemented moment vector is

$$g_i(\beta) = \begin{pmatrix} z_i^c(y_i^c - \beta x_i^c) \\ \\ \widetilde{x}_i(\widetilde{y}_i - \beta \widetilde{x}_i) \end{pmatrix}.$$

IV and SV use the first and second moment conditions, respectively. Two-step GMM uses identity weighting in the first step and the generalized inverse of the batch-clustered moment covariance evaluated at the first-step estimate in the second step. Standard errors for all estimators are clustered by batch.

For local misspecification-adjusted inference, we evaluate $\widehat{g}_i = g_i(\widehat{\beta}_{\mathrm{GMM}})$ and compute

$$\widehat{\Omega} = \frac{G}{G-1}\frac{1}{n}\sum_{b=1}^{G}\left\{\sum_{i\in\mathcal{I}_b}(\widehat{g}_i - \bar{g})\right\}\left\{\sum_{i\in\mathcal{I}_b}(\widehat{g}_i - \bar{g})\right\}^{\top},$$

where $G = 48$, $\mathcal{I}_b$ indexes profiles in batch b, and $\bar{g} = n^{-1}\sum_i \widehat{g}_i$. We set $r_j = 0.5\sqrt{\widehat{\Omega}_{jj}}$ and use the same ellipsoidal deviation set for IV, SV, and GMM. The adjusted inference accounts for the worst-case local bias over this set while retaining each method's point estimate.

Multiple testing and biological enrichment. For each perturbation, we apply Benjamini–Hochberg adjustment across the 2,000 outcomes separately for each estimator and each inference rule. For the Wilcoxon comparison, raw p-values are readjusted over the same 2,000 outcomes within each perturbation. GO biological-process enrichment uses the tested outcome panel as its background: 1,790 of the 2,000 genes have direct GO:BP annotations, and 4,795 terms are tested for each perturbation-specific discovery set.